



**ROYAL SCHOOL OF BIO-SCIENCES
(RSBSC)**

DEPARTMENT OF BIOCHEMISTRY

**COURSE STRUCTURE & SYLLABUS
(BASED ON NATIONAL EDUCATION POLICY 2020)
FOR
B.SC. IN BIOCHEMISTRY
(4 YEARS SINGLE MAJOR)**

**WEF
AY- 2023- 2024**

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1. PREAMBLE

The National Education Policy (NEP) 2020 conceives a new vision for India's higher education system. It recognizes that higher education plays an extremely important role in promoting equity, human as well as societal well-being and in developing India as envisioned in its Constitution. It is desired that higher education will significantly contribute towards sustainable livelihoods and economic development of the nation as India moves towards becoming a knowledge economy and society.

If we focus on the 21st century requirements, the higher education framework of the nation must aim to develop good, thoughtful, well-rounded, and creative individuals and must enable an individual to study one or more specialized areas of interest at a deep level, and also develop character, ethical and Constitutional values, intellectual curiosity, scientific temper, creativity, spirit of service, and twenty-first-century capabilities across a range of disciplines including sciences, social sciences, arts, humanities, languages, as well as professional, technical, and vocational subjects. A quality higher education should be capable enough to enable personal accomplishment and enlightenment, constructive public engagement, and productive contribution to the society. Overall, it should focus on preparing students for more meaningful and satisfying lives and work roles and enable economic independence.

Towards the attainment of holistic and multidisciplinary education, the flexible curricula of the University will include credit-based courses, projects in the areas of community engagement and service, environmental education, and value-based education. As part of holistic education, students will also be provided with opportunities for internships with local industries, businesses, artists, crafts persons, and so on, as well as research internships with faculty and researchers at the University, so that students may actively engage with the practical aspects of their learning and thereby improve their employability.

The undergraduate curriculums are diverse and have varied subjects to be covered to meet the needs of the programs. As per the recommendations from the UGC, introduction of courses related to Indian Knowledge System (IKS) is being incorporated in the curriculum structure which encompasses all of the systematized disciplines of Knowledge which were developed to a high degree of sophistication in India from ancient times and all of the traditions and practices that the various communities of India—including the tribal communities—have evolved, refined and preserved over generations, like for example Vedic Mathematics, Vedangas, Indian Astronomy, Fine Arts, Metallurgy, etc.

At RGU, we are committed that at the societal level, higher education will enable each student to develop themselves to be an enlightened, socially conscious, knowledgeable, and skilled citizen who can find and implement robust solutions to its own problems. For the students at the University, Higher education is expected to form the basis for knowledge creation and innovation thereby contributing to a more vibrant, socially engaged, cooperative community leading towards a happier, cohesive, cultured, productive, innovative, progressive, and prosperous nation.”

The new curriculum of BSc in Biochemistry under The Assam Royal Global University will be—more flexible, multi-disciplinary and holistic. Overall, the revamped BSc in Biochemistry curriculum at The Assam Royal Global University will empower students with a well-rounded education, preparing them to tackle the dynamic and complex challenges of the modern world while fostering a lifelong passion for biological sciences.

2. INTRODUCTION

The National Education Policy (NEP) 2020 clearly indicates that higher education plays an extremely important role in promoting human as well as societal well-being in India. As envisioned in the 21st-century requirements, quality higher education must aim to develop good, thoughtful, well-rounded, and creative individuals. According to the new education policy, assessments of educational approaches in undergraduate education will integrate the humanities and arts with Science, Technology, Engineering and Mathematics (STEM) that will lead to positive learning outcomes. This will lead to develop creativity and innovation, critical thinking and higher-order thinking capacities, problem-solving abilities, teamwork, communication skills, more in-depth learning, and mastery of curricula across fields, increases in social and moral awareness, etc., besides general engagement and enjoyment of learning. and more in-depth learning.

The NEP highlights that the following fundamental principles that have a direct bearing on the curricula would guide the education system at large, viz.

- i. Recognizing, identifying, and fostering the unique capabilities of each student to promote her/his holistic development.
- ii. Flexibility, so that learners can select their learning trajectories and programmes, and thereby choose their own paths in life according to their talents and interests.
- iii. Multidisciplinary and holistic education across the sciences, social sciences, arts, humanities, and sports for a multidisciplinary world.
- iv. Emphasis on conceptual understanding rather than rote learning, critical thinking to encourage logical decision-making and innovation; ethics and human & constitutional values, and life skills such as communication, teamwork, leadership, and resilience.
- v. Extensive use of technology in teaching and learning, removing language barriers, increasing access for Divyang students, and educational planning and management.
- vi. Respect for diversity and respect for the local context in all curricula, pedagogy, and policy.
- vii. Equity and inclusion as the cornerstone of all educational decisions to ensure that all

students can thrive in the education system and the institutional environment are responsive to differences to ensure that high-quality education is available for all.

- viii. Rootedness and pride in India, and its rich, diverse, ancient, and modern culture, languages, knowledge systems, and traditions.

2.1 Credits in Indian Context:

Choice Based Credit System (CBCS) By UGC

Under the CBCS system, the requirement for awarding a degree or diploma or certificate is prescribed in terms of number of credits to be earned by the students. This framework is being implemented in several universities across States in India. The main highlights of CBCS are as below:

- The CBCS provides flexibility in designing curriculum and assigning credits based on the course content and learning hours.
- The CBCS provides for a system wherein students can take courses of their choice, learn at their own pace, undergo additional courses and acquire more than the required credits, and adopt an interdisciplinary approach to learning.
- CBCS also provides opportunity for vertical mobility to students from a bachelor's degree programme to masters and research degree programmes.

2.2 Definitions

2.2.1 Academic Credit:

An academic credit is a unit by which a course is weighted. It is fixed by the number of hours of instructions offered per week. As per the National Credit Framework [2];

1 Credit = 30 NOTIONAL CREDIT HOURS (NCH)

Yearly Learning Hours = 1200 Notional Hours (@40 Credits x 30 NCH)

30NotionalCreditHours		
Lecture/Tutorial	Practicum	Experiential Learning

1Credit=15-22 Lecture Hours	10-15 Practicum Hours	0-8 Experiential Learning Hours
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Looking at all these new concepts and progress, the detailed syllabus of B.Sc. in Biochemistry has been designed and decided to be implemented. Biochemistry deals with the structure and function of biomolecules and the vital processes that occur in living organisms including plants, animals and microorganisms. Biochemistry has application in clinical diagnosis, understanding pathology of diseases and treatment options, drug designing, food adulteration, and understanding the metabolism of various biological products like amino acids, proteins, antibiotics, hormones, nutrients, enzymes etc. As it covers a diverse range of topics, modern biochemists need to have insight into many disciplines. The learning outcomes-based curriculum framework for a B.Sc. in Biochemistry is designed to cater to the needs of students in view of the evolving nature of animal science as a subject. The present framework is intended to allow for flexibility and innovation in programme design and syllabi development, teaching-learning process, assessment of student learning levels.

3. APPROACH TO CURRICULUM PLANNING

The fundamental premise underlying the learning outcomes-based approach to curriculum planning and development is that higher education qualifications such as a Bachelor's programmes are earned and awarded on the basis of (a) demonstrated achievement of outcomes (expressed in terms of knowledge, understanding, skills, attitudes and values) and (b) academic standards expected of graduates of a programme of study.

The expected learning outcomes are used as reference points that would help formulate graduate attributes, qualification descriptors, programme learning outcomes and course learning outcomes which in turn will help in curriculum planning and development, and in the design, delivery, and review of academic programmes. Learning outcomes-based frameworks in any subject must specify what graduates completing a particular programme of study are (a) expected to know, (b) understand and (c) be able to do at the end of their programme of study. To this extent, the course content of Biochemistry is committed to allowing for flexibility and innovation in (i) programme design and syllabi development by higher education institutions (HEIs), (ii) teaching-learning process, (iii) assessment of student learning levels, and (iv) periodic

programme review within institutional parameters (v) generating framework(s) of agreed expected graduate attributes, qualification descriptors, programme learning outcomes and course learning outcomes.

The key outcomes that underpin curriculum planning and development at the undergraduate level include Graduate Attributes, Qualification Descriptors, Programme Learning Outcomes, and Course Learning Outcomes. The undergraduate education is based on specific learning outcomes and academic standards expected to be attained by graduates of a programme of study. However, an outcome-based approach identifies moves away from the emphasis on what is to be taught to focus on what is learnt by way of demonstrable outcomes. This approach provides greater flexibility to the teachers to develop—and the students to accept and adopt—different learning and teaching pedagogy in an interactive and participatory ecosystem. The idea is to integrate social needs and teaching practices in a manner that is responsive to the need of the community. HEIs, on their turn, shall address to the situations of their students by identifying relevant and common outcomes and by developing such outcomes that not only match the specific needs of the students but also expands their outlook and values.

B.Sc. in Biochemistry is a recognized and structured undergraduate degree in tertiary education. It offers specialized knowledge, skills, and values that students can use to pursue professional opportunities or continue their studies at the postgraduate level. The purpose of the present curriculum of B.Sc. Biochemistry is to prepare students for professional careers or further studies in various job fields. Graduates can pursue a wide range of jobs or continue their education at an advanced level.

3.1 Course of Study:

Course of study indicate pursuance of study in a particular discipline/programme. Discipline/Programmes shall offer Major Courses (Core), Minor Courses, Skill Enhancement Courses (SEC), Value Added Courses (VAC), Ability Enhancement Compulsory Courses (AECCs) and Interdisciplinary courses.

3.2 Disciplinary Major:

The major would provide the opportunity for a student to pursue in-depth study of a particular

subject or discipline. Students may be allowed to change major within the broad discipline at the end of the second Semester by giving her/him sufficient time to explore interdisciplinary courses during the first year. Advanced-level disciplinary/interdisciplinary courses, a course in research methodology, and a project/dissertation will be conducted in the seventh Semester. The final Semester will be devoted to seminar presentation, preparation, and submission of project report/dissertation. The project work/dissertation will be on a topic in the disciplinary programme of study or an interdisciplinary topic.

3.3 Disciplinary/Interdisciplinary Minors:

Students will have the option to choose courses from disciplinary/interdisciplinary minors and skill-based courses. Students who take a sufficient number of courses in a discipline or an interdisciplinary area of study other than the chosen major will qualify for a minor in that discipline or in the chosen interdisciplinary area of study. A student may declare the choice of the minor at the end of the second Semester, after exploring various courses.

3.4 Courses from Other Disciplines (Interdisciplinary):

All UG students are required to undergo 3 introductory-level courses relating to any of the broad disciplines given below. These courses are intended to broaden the intellectual experience and form part of liberal arts and science education. Students are not allowed to choose or repeat courses already undergone at the higher secondary level (12th class) in the proposed major and minor stream under this category.

- ***Natural and Physical Sciences:*** Students can choose basic courses from disciplines such as Natural Science, for example, Biology, Botany, Zoology, Biotechnology, Biochemistry, Chemistry, Physics, Biophysics, Astronomy and Astrophysics, Earth and Environmental Sciences, etc.
- ***Mathematics, Statistics, and Computer Applications:*** Courses under this category will facilitate the students to use and apply tools and techniques in their major and minor disciplines. The course may include training in programming software like Python among others and applications software like STATA, SPSS, Tally, etc. Basic courses under this category will be helpful for science and social science in data analysis and the application

of quantitative tools.

- ***Library, Information, and Media Sciences:*** Courses from this category will help the students to understand the recent developments in information and media science (journalism, mass media, and communication)
- ***Commerce and Management:*** Courses include business management, accountancy, finance, financial institutions, fintech, etc.
- ***Humanities and Social Sciences:*** The courses relating to Social Sciences, for example, Anthropology, Communication and Media, Economics, History, Linguistics, Political Science, Psychology, Social Work, Sociology, etc. will enable students to understand the individuals and their social behaviour, society, and nation. Students shall be introduced to survey methodology and available large-scale databases for India. The courses under humanities include, for example, Archaeology, History, Comparative Literature, Arts & Creative expressions, Creative Writing and Literature, language(s), Philosophy, etc., and interdisciplinary courses relating to humanities. The list of Courses can include interdisciplinary subjects such as Cognitive Science, Environmental Science, Gender Studies, Global Environment & Health, International Relations, Political Economy and Development, Sustainable Development, Women's, and Gender Studies, etc. will be useful to understand society.

3.5 Ability Enhancement Courses (AEC):

Modern Indian Language (MIL) & English language focused on language and communication skills. Students are required to achieve competency in a Modern Indian Language (MIL) and in the English language with special emphasis on language and communication skills. The courses aim at enabling the students to acquire and demonstrate the core linguistic skills, including critical reading and expository and academic writing skills, that help students articulate their arguments and present their thinking clearly and coherently and recognize the importance of language as a mediator of knowledge and identity. They would also enable students to acquaint themselves with the cultural and intellectual heritage of the chosen MIL and English language, as well as to provide a reflective understanding of the structure and complexity of the language/literature related to both the MIL and English language. The courses will also

emphasize the development and enhancement of skills such as communication, and the ability to participate/conduct discussion and debate.

3.6. Skill Enhancement Course (SEC):

These courses are aimed at imparting practical skills, hands-on training, soft skills, etc., to enhance the employability of students and should be related to Major Discipline. They will aim at providing hands-on training, competencies, proficiency, and skill to students. SEC course will be a basket course to provide skill-based instruction. For example, SEC of English Discipline may include Public Speaking, Translation & Editing and Content writing. A student shall have the choice to choose from a list, a defined track of courses offered from 1st to 3rd Semester.

3.7. Value-Added Courses(VAC):

- ***Understanding India:*** The course aims at enabling the students to acquire and demonstrate the knowledge and understanding of contemporary India with its historical perspective, the basic framework of the goals and policies of national development, and the constitutional obligations with special emphasis on constitutional values and fundamental rights and duties. The course would also focus on developing an understanding among student-teachers of the Indian knowledge systems, the Indian education system, and the roles and obligations of teachers to the nation in general and to the school/community/society. The course will attempt to deepen knowledge about and understanding of India's freedom struggle and of the values and ideals that it represented to develop an appreciation of the contributions made by people of all sections and regions of the country, and help learners understand and cherish the values enshrined in the Indian Constitution and to prepare them for their roles and responsibilities as effective citizens of a democratic society.
- ***Environmental science/education:*** The course seeks to equip students with the ability to apply the acquired knowledge, skills, attitudes, and values required to take appropriate actions for mitigating the effects of environmental degradation, climate change, and pollution, effective waste management, conservation of biological diversity, management of biological resources, forest and wildlife conservation, and sustainable development and living. The course will also deepen the knowledge and understanding of India's

environment in its totality, its interactive processes, and its effects on the future quality of people's lives.

- **Digital and technological solutions:** Courses in cutting-edge areas that are fast gaining prominences, such as Artificial Intelligence (AI), 3-D machining, big data analysis, machine learning, drone technologies, and Deep learning with important applications to health, environment, and sustainable living that will be woven into undergraduate education for enhancing the employability of the youth.
- **Health & Wellness, Yoga education, sports, and fitness:** Course components relating to health and wellness seek to promote an optimal state of physical, emotional, intellectual, social, spiritual, and environmental well-being of a person. Sports and fitness activities will be organized outside the regular institutional working hours. Yoga education would focus on preparing the students physically and mentally for the integration of their physical, mental, and spiritual faculties, and equipping them with basic knowledge about one's personality, maintaining self-discipline and self-control, to learn to handle oneself well in all life situations. The focus of sports and fitness components of the courses will be on the improvement of physical fitness including the improvement of various components of physical and skills-related fitness like strength, speed, coordination, endurance, and flexibility; acquisition of sports skills including motor skills as well as basic movement skills relevant to a particular sport; improvement of tactical abilities; and improvement of mental abilities. These are a common pool of courses offered by different disciplines and aimed towards embedding ethical, cultural and constitutional values; promote critical thinking. Indian knowledge systems; scientific temperament of students.

3.8 Summer Internship /Apprenticeship:

The intention is induction into actual work situations. All students must undergo internships / Apprenticeships in a firm, industry, or organization or Training in labs with faculty and researchers in their own or other HEIs/research institutions during the summer term. Students should take up opportunities for internships with local industry, business organizations, health and allied areas, local governments (such as panchayats, municipalities), Parliament or elected representatives, media organizations, artists, crafts persons, and a wide variety of organizations

so that students may actively engage with the practical side of their learning and, as a by-product, further improve their employability. Students who wish to exit after the first two semesters will undergo a 4-credit work-based learning/internship during the summer term to get a UG Certificate.

3.9 Community engagement and service:

The curricular component of ‘community engagement and service’ seeks to expose students to the socio- economic issues in society so that the theoretical learnings can be supplemented by actual life experiences to generate solutions to real-life problems. This can be part of summer term activity or part of a major or minor course depending upon the major discipline.

3.10 Field-based learning/minor project:

The field-based learning/minor project will attempt to provide opportunities for students to understand the different socio-economic contexts. It will aim at giving students exposure to development-related issues in rural and urban settings. It will provide opportunities for students to observe situations in rural and urban contexts, and to observe and study actual field situations regarding issues related to socioeconomic development. Students will be given opportunities to gain a first- hand understanding of the policies, regulations, organizational structures, processes, and programmes that guide the development process. They would have the opportunity to gain an understanding of the complex socio-economic problems in the community, and innovative practices required to generate solutions to identified problems. This may be as summer term project or part of a major or minor course depending on the subject of study.

3.11 Indian Knowledge System:

In view of the importance accorded in the NEP 2020 to rooting our curricula and pedagogy in the Indian context all the students who are enrolled in the four-year UG programmes should be encouraged to take an adequate number of courses in IKS so that the ***total credits of the courses taken in IKS amount to at least five per cent of the total mandated credits (i.e. min. 8credits for a 4 yr. UGP & 6 credits for a 3 yr. UGP).*** The students may be encouraged to take these courses, preferably *during the first four semesters of the UG programme.* At least half of these mandated credits should be in courses in disciplines which are part of IKS and are related to the major field

of specialization that the student is pursuing in the UG programme. They will be included as a part of the total mandated credits that the student is expected to take in the major field of specialization. The rest of the mandated credits in IKS can be included as a part of the mandated Multidisciplinary courses that are to be taken by every student. All the students should take a Foundational Course in Indian Knowledge System, which is designed to present an overall introduction to all the streams of IKS relevant to the UG programme. The foundational IKS courses should be broad-based and cover introductory material on all aspects. Wherever possible, the students may be encouraged to choose a suitable topic related to IKS for their project work in the 7/8th semesters of the UG programme.

3.12 Experiential Learning:

One of the most unique, practical & beneficial features of the National Credit Framework is assignment of credits/credit points/ weightage to the experiential learning including relevant experience and professional levels acquired/ proficiency/ professional levels of a learner/student. Experiential learning is of two types:

- ***Experiential learning as part of the curricular structure*** of academic or vocational program. E.g., projects/OJT/internship/industrial attachments etc. This could be either within the Program- internship/ summer project undertaken relevant to the program being studied or as a part time employment (not relevant to the program being studied-up to certain NSQF level only). In case where experiential learning is a part of the curricular structure the credits would be calculated and assigned as per basic principles of NCrF i.e., 40 credits for 1200 hours of notional learning.
- ***Experiential learning as active employment*** (both wage and self) post completion of an academic or vocational program. This means that the experience attained by a person after undergoing a particular educational program shall be considered for assignment of credits. This could be either Full or Part time employment after undertaking an academic/ Vocation program.

In case where experiential learning is as a part of employment the learner would earn credits as weightage. The maximum credit points earned in this case shall be double of the credit points earned with respect to the qualification/ course completed. The credit earned and assigned by virtue of relevant experience would enable learners to progress in their career through the work

hours put in during a job/employment.

4. AWARD OF DEGREE

The structure and duration of undergraduate programmes of study offered by the University as per NEP 2020 include:

Undergraduate programmes of either 3 or 4-year duration with a Single Major, with multiple entry and exit options, with appropriate certifications:

UG Certificate: Students who opt to exit after completion of the first year and have secured 40 credits will be awarded a UG certificate if, in addition, they complete one vocational course of 4 credits during the summer vacation of the first year. These students are allowed to re-enter the degree programme within three years and complete the degree programme within the stipulated maximum period of seven years.

UG Diploma: Students who opt to exit after completion of the second year and have secured 80 credits will be awarded the UG diploma if, in addition, they complete one vocational course of 4 credits during the summer vacation of the second year. These students are allowed to re-enter within a period of three years and complete the degree programme within the maximum period of seven years.

3-year UG Degree: Students who will undergo a 3-year UG programme will be awarded UG Degree in the Major discipline after successful completion of three years, securing 120 credits and satisfying the minimum credit requirement.

4-year UG Degree (Honours): A four-year UG Honours degree in the major discipline will be awarded to those who complete a four-year degree programme with 160 credits and have satisfied the credit requirements as given in Table 6 in Section 5.

4-year UG Degree (Honours with Research): Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year. They should do a research project or dissertation under the guidance of a Faculty Member of the University. The research project/dissertation will be in the major discipline. The students who secure 160 credits, including 12 credits from a research project/dissertation, will be awarded UG Degree (Honours with Research).

(Note: **UG Degree Programmes with Single Major:** A student must secure a minimum of 50% credits from the major discipline for the 3-year/4-year UG degree to be awarded a single major. For example, in a 3-year UG programme, if the total number of credits to be earned is 120, a student of Mathematics with a minimum of 60 credits will be awarded a B.Sc. in Mathematics with a single major. Similarly, in a 4-year UG programme, if the total number of credits to be earned is 160, a student of Chemistry with a minimum of 80 credits will be awarded a B.Sc. (Hons. /Hon. With Research) in Chemistry in a 4-year UG programme with single major. Also, the **4-year Bachelor's degree programme with Single Major** is considered as the preferred option since it would allow the opportunity to experience the full range of holistic and multidisciplinary education in addition to a focus on the chosen major and minors as per the choices of the student.)

award	year	credits to earn	additional credits	re-entry allowed within(yrs)	years to complete
UG certificate	1	40	4	3	7
UG diploma	2	80	4	3	7
3-year UG degree(major)	3	120	x	x	x
4-year UG degree (honors)	4	160	x	x	x
Award	Year	Credits to earn	Additional credits	Re-entry allowed within(yrs)	Years to complete
4-year UG Degree (Honors with Research):	4	160	Students who secure cumulative 75% marks and above in the first six semesters		

5. GRADUATE ATTRIBUTES & LEARNING OUTCOMES

As per the NHEQF, each student on completion of a programme of study must possess and demonstrate the expected **Graduate Attributes** acquired through one or more modes of learning, including direct in-person or face-to-face instruction, online learning, and hybrid/blended modes. The graduate attributes indicate the quality and features or characteristics of the graduate of a

programme of study, including learning outcomes relating to the disciplinary area(s) relating to the chosen field(s) of learning and generic learning outcomes that are expected to be acquired by a graduate on completion of the programme(s) of study.

The graduate profile/attributes are acquired incrementally through development of cognitive levels and describe a set of competencies that are transferable beyond the study of a particular subject/disciplinary area and programme contexts in which they have been developed.

Graduate attributes include,

- ***learning outcomes that are specific to disciplinary areas*** relating to the chosen field(s) of learning within broad multidisciplinary/interdisciplinary/ transdisciplinary contexts.
- ***generic learning outcomes*** that graduate of all programmes of study should acquire and demonstrate.

Table: The Learning Outcomes Descriptors and Graduate Attributes		
Sl.no.	Graduate Attribute	The Learning Outcomes Descriptors (<i>The graduates should be able to demonstrate the capability to:</i>)
GA1	Disciplinary Knowledge	Acquire knowledge and coherent understanding of the chosen disciplinary/interdisciplinary areas of study.
GA 2	Complex problem solving	Solve different kinds of problems in familiar and non familiar contexts and apply the learning to real-life situations.
GA 3	Analytical & Critical thinking	Apply analytical thought including the analysis and evaluation of policies, and practices. Able to identify relevant assumptions or implications.
GA 4	Creativity	Create, perform, or think indifferent and diverse ways about the same objects or scenarios and deal with problems and situations that do not have simple solutions. Think ‘out of the box’ and generate solutions to complex problems in unfamiliar contexts by adopting innovative, imaginative, lateral thinking, interpersonal skills, and emotional intelligence.

GA 5	Communication Skills	Listen carefully, read texts and research papers analytically, and present complex information in a clear and concise manner to different groups/audiences. Express thoughts and ideas effectively in writing and orally and communicate with others using appropriate media.
GA 6	Research-related skills	Develop a keen sense of observation, inquiry, and capability for asking relevant/ appropriate questions. Should acquire the ability to problematize, synthesize and articulate issues and design research proposals, define problems, formulate appropriate and relevant research questions, formulate hypotheses, test hypotheses using quantitative and qualitative data, establish hypotheses, make inferences based on the analysis and interpretation of data, and predict cause-and-effect relationships. Should develop the ability to acquire the understanding of basic research ethics and skills in practicing/doing ethics in the field/ in personal research work.
GA 7	Collaboration	Work effectively and respect fully with diverse teams in the interests of a common cause and work efficiently as a member of a team.
GA 8	Leadership readiness/qualities	Plan the tasks of a team or an organization and setting direction by formulating an inspiring vision and building a team that can help achieve the vision.
GA 9	Digital and technological skills	Use ICT in a variety of learning and work situations. Access, evaluate, and use a variety of relevant information sources and use appropriate software for analysis of data.
GA 10	Environmental awareness and action	Mitigate the effects of environmental degradation, climate change, and pollution. Should develop the technique of effective waste management, conservation of biological diversity, management of biological resources and biodiversity, forest and wildlife conservation, and sustainable development and living.

6. PROGRAMME LEARNING OUTCOMES

PO-1: Knowledge of Biochemistry

Upon completion of the Biochemistry program, graduates will demonstrate the ability to apply fundamental knowledge of Biomolecules, protein, biochemical techniques in the area of Biochemistry.

PO-2:Ability to solve Complex problems

Upon completion of the Biochemistry program, graduates will possess advanced skills in complex problem-solving, enabling them to address intricate challenges in biochemistry and allied sciences.

PO-3:Develop analytical and critical thinking

Graduates will exhibit advanced capabilities in analytical and critical thinking, allowing them to assess scientific data, conduct rigorous research, and evaluate complex biological issues with precision and insight.

PO-4:Develop the ability to create

Graduates will demonstrate innovative and creative thinking, fostering novel approaches to address biological challenges and contribute to advancements in biochemistry research and innovation. They will harness the ability to learn a system with its component, or process to meet desired need within realistic constraints.

PO-5:Develop effective communication skills

Graduates will possess strong communication skills, effectively conveying complex scientific concepts and findings to diverse audiences, fostering collaboration and promoting public awareness of biological issues.

PO-6:Develop research related skills

Upon completion of the Biochemistry program, graduates will possess the ability to use the techniques, skills and modern professional tools necessary for professional practice and for research.

PO-7:Develop skills for Collaborative/Team Work

The students will exhibit strong collaboration skills, effectively working with multidisciplinary teams to address complex challenges and promote collective efforts towards biochemistry research and its application in human welfare.

PO-8:Develop Leadership qualities

Upon completion of Biochemistry program, graduates will demonstrate exceptional leadership qualities, inspiring and guiding teams in the field of biochemistry and allied sciences to foster innovation, conservation, and sustainable practices.

PO-9:Enable students to adopt to evolving digital and technological skills

Graduates will possess proficient digital and technological skills, utilizing cutting-edge tools and methodologies to enhance research, data analysis, and communication in the field of biochemistry, biotechnology, and bioinformatics, promoting advancements and efficiency in their work.

PO-10:Develop environmental awareness and develop suitable skills to address to problems

Upon completion of the program, graduates will display a heightened environmental awareness, incorporating ecological principles into their work, and taking proactive actions to promote sustainability and conservation efforts in the realm of biology.

7. PROGRAMME SPECIFIC OUTCOMES

PSO1: Ability to apply knowledge of Biochemistry to realize and explain notions and complexities of biological and allied sciences.

PSO2: Ability to comprehend the relationship of organisms at all levels: molecular, cellular, and organismal and students shall be able to conduct the clinical biochemistry, diagnostic biochemistry experiments as well as to analyze and interpret the results.

PSO3:Students shall be able to use the biochemical techniques, genetic engineering & biotechnology skills and modern pathological tools necessary for professional practice and conduct empirical studies for scientific research as well as to validate, analyze and interpret

them.

8. TEACHING LEARNING PROCESS

Teaching and learning in this programme involve classroom lectures, tutorials, flipped classrooms, seminars/presentations, hands-on practical, and in-housed and out-housed internships and research projects.

The various aspects are shown below:

- Written assignments and projects submitted by students
- Project-based learning, group discussion, home assignments, quizzes and class tests
- Reflective learning and flipped classrooms
- PPT presentations, seminars, interactive sessions, co-curricular activities, etc.
- Industrial Tour or Field visit

9. ASSESSMENT METHODS

Methods	Weightage
Semester End Examination	70%
Internal Assessment	30%
Total	100%

Internal assessment is based on:

- 25%- Mid-semester Examination, Class test, Assignment, Project, Viva-voce, Seminar, etc.
- 5%- Attendance of the student.

STRUCTURE OF THE SYLLABUS FOR 4 YEAR UG PROGRAMME

SCHOOL NAME - Royal School of Biosciences

DEPARTMENT NAME - Biochemistry

PROGRAMME NAME - UG Biochemistry

1ST SEMESTER

COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-C
Major (Core)	BCH152M101	Biomolecules	100	3	3-0-0
Major	BCH152M111	Biomolecules and Cell Biology Practical	100	3	0-0-6
Minor	BCH152N101	Nutritional Biochemistry	100	3	3-0-0
Interdisciplinary (IDC)	IKS992K201	IKS-1	100	3	3-0-0
Ability Enhancement course (AEC)	AEC982A101	Communicative English-I	100	1	1-0-0
AEC	BHS982A101	Behavioral Science I	100	1	1-0-0
Skill Enhancement Course (SEC)	BCH152S111	Biochemical Assessment of Food Products	100	3	0-0-6
Value Added Course (VAC)		To be chosen from basket	100	3	3-0-0
*MOOCs	MOOCs 1	*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online	100	3	

		platform/SWAYAM portal			
TOTAL CREDIT FOR 1st SEMESTER				23	
2ND SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-C
Major (Core)	BCH152M201	Proteins and Enzymes	100	3	3-0-0
Major (Core)	BCH152M213	Protein and Enzymes Practical	100	3	0-0-6
Minor	BCH152N201	Introduction to Cytology	100	3	3-0-0
IDC	IKS992K202	IKS-2		3	3-0-0
AEC	AEC982A201	Communicative English- II	100	1	1-0-0
AEC	BHS982A201	Behavioral Science II	100	1	1-0-0
SEC	BCH152S211	Food Adulteration	100	3	0-0-6
VAC		To be chosen from Basket		3	3-0-0
*MOOCs		*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	
TOTAL CREDIT FOR 2nd SEMESTER				23	

3 RD SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-C
Major (Core)	BCH152M301	Metabolism of Biomolecules	200	4	3-1-0
Major (Core)	BCH152M313	Metabolism Practical	200	4	0-0-6
Minor	BCH152N301	Clinical Biochemistry	200	4	3-0-0
IDC	BCH152I301	Immunity and Health	200	3	3-0-0
AEC	AEC982A301	Communicative English-III	200	1	1-0-0
AEC	BHS982A301	Behavioral Science III	200	1	1-0-0
SEC	BCH152S311	Working with proteins	200	3	0-0-6
*MOOCs		*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	
TOTAL CREDIT FOR 3 rd SEMESTER				23	
	4 th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M401	Concepts of Genetics	200	4	3-1-0

Major (Core)	BCH152M402	Human diseases and Basic Ayurveda	200	4	3-1-0
Major (Core)	BCH152M413	Practical-IV (Genetics)	200	4	0-0-6
Minor	BCH152N401	Microbiology	200	3	3-0-0
Minor	BCH152N402	Biochemical Correlation of Diseases	200	3	3-0-0
AEC	AEC982A401	Communicative English-IV	200	1	1-0-0
AEC	BHS982A401	Behavioral Science IV	200	1	1-0-0
*MOOCs		*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	
TOTAL CREDIT FOR 4th SEMESTER				23	
	5th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M501	Molecular Biology	300	4	3-1-0
Major (Core)	BCH152M502	Microbiology	300	4	3-1-0
Major (Core)	BCH152M513	Molecular Biology and Microbiology Practical	300	4	0-0-8
Minor	BCH152N501	Environmental	300	4	3-1-0

		Biochemistry			
Internship	BCH152M522	Internship		4	
TOTAL CREDIT FOR 5th SEMESTER				20	
	6th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M601	Advanced Cell Biology	300	4	3-1-0
Major (Core)	BCH152M602	Immunology	300	4	3-1-0
Major (Core)	BCH152M603	Membrane Biology and Bioenergetics	300	4	3-1-0
Major (Core)	BCH152M613	Immunology and Cell Biology Practical	300	4	0-0-8
Minor	BCH152N601	Basics of Genetic Engineering	200	4	3-1-0
TOTAL CREDIT FOR 6th SEMESTER				20	
	7th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M701	Genetic Engineering and Biotechnology	400	4	3-1-0
Major (Core)	BCH152M702	Research Methodology, Bioethics, and Scientific Writing	400	4	3-1-0
Major (Core)	BCH152M703	Plant Biochemistry	400	4	3-1-0

Major (Core)	BCH152M713	Plant Biology and Genetic Engineering Practical	400	4	3-1-0
Minor	BCH152N701	Cancer Biology	400	4	3-1-0
TOTAL CREDIT FOR 7th SEMESTER				20	
	8th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M801	Bioanalytical techniques	400	4	3-1-0
Minor	BCH152N801	Microbial Ecology	400	4	3-1-0
		Project / Dissertation		12	
		Or			
		Virology		4	3-1-0
		Neurobiochemistry		4	3-1-0
		Molecular Endocrinology		4	3-1-0
TOTAL CREDIT FOR 8th SEMESTER				20	

SYLLABUS 1st SEMESTER

MAJOR PAPER: Biomolecules

Course level: 100

Subject Code:

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

The course aims to provide students with an understanding of biomolecules, the basic building blocks of living organisms, focusing on their structural underpinnings, unique properties, biological roles and functions, and interrelations.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Define and distinguish the different biomolecules in a living system	BT1 & BT2
CO 2	Summarize the importance of each biomolecules-viz. nucleic acid, carbohydrate, lipid, and protein along with vitamins and water.	BT2

CO 3	Illustrate and sketch the chemical structures of major biomolecules and their building blocks.	BT3
CO 4	Apply the knowledge to analyze the critical role of biomolecules as the foundation of living organisms	BT4

Course Contents

Modules	Course content	Periods
I	The foundations of biochemistry: Cellular and chemical foundations of life. Water: Unique properties, weak interactions in aqueous systems, ionization of water, buffers, water as a reactant, and fitness of the aqueous environment. Vitamins: Structure and active forms of water-soluble and fat-soluble vitamins, deficiency diseases and symptoms, hypervitaminosis	15
II	Carbohydrates and glycobiology: Monosaccharides –the structure of aldoses and ketoses, the ring structure of sugars, conformations of sugars, mutarotation, anomers, epimers and enantiomers, the structure of biologically important sugar derivatives, oxidation of sugars. Disaccharides (examples of disaccharides). Polysaccharides homo- and heteropolysaccharides, structural and storage polysaccharides. Structure and role of proteoglycans, glycoproteins, and glycolipids (gangliosides and lipopolysaccharides). Carbohydrates as informational molecules, working with carbohydrates	15
III	Lipids: Building blocks of lipids - fatty acids, glycerol, and ceramide. Storage lipids - triacyl glycerol and waxes. Structural lipids in membranes – glycerophospholipids, galactolipids and sulpholipids, sphingolipids and sterols, structure, distribution, and role of membrane lipids.	15
IV	Amino acids and Proteins: Structure and classification, physical, chemical and optical properties of amino acids (hydrophobic, polar and charged). Proteins and peptide bonds. Nucleic acids: Nucleotides - structure and properties. Nucleic acid structure – Watson-Crick model of DNA. Structure of major species of RNA - mRNA, tRNA, and rRNA. Nucleic acid chemistry- UV absorption, the effect of acid and alkali on DNA.	15
	Total	60

Textbooks:

Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company,

New York, USA, 7th edition, 2017.

Victor Rodwell, David Bender, P. Anthony Weil, Peter Kennelly. Harpers Illustrated Biochemistry 31th Edition, 2018.

Reference books:

Principles of Biochemistry, 5th Edition. 2011, Robert Horton H, Laurenc Moran, Gray Scrimgeour K. Pearsarson Publisher. ISBN 978-0321707338.

Introductory Practical Biochemistry by S.K. Sawhney and R. Singh, 4th Edition, Alpha Science International, 2017. ISBN-10:1842652451.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom

MAJOR PAPER: Practical I (Biomolecules Practical)

Course level: 100

Subject Code:

L-T-P-C: 0-0-6-3

Scheme of evaluation: Practical (P)

Course Objective

The objective of this paper is to offer insights about the practicals based on analysis of biomolecules as well on the cell biology.

Course Outcome

On Successful Completion of the course, the students will be able to:
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Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Identify and recall basic laboratory equipment and technique commonly used in Biochemistry and relate the understanding of biomolecules to cell biology.	BT1
CO 2	Describe and demonstrate various buffers preparation, buffering range, its uses and application in biochemical reactions, enzyme activity and blood group variety testing etc.	BT2& BT3
CO 3	Apply the knowledge of cell biology in practical and interpret experimental results.	BT3
CO 4	Illustrate the process cell division practically and develop practical skills in cell biology.	BT4

Course Contents

Modules	Course content	Periods
I	1. Basic Lab requirements: Volumetric flask, falcons, mortar and pestle, watch glass, wash bottle, beaker, measuring cylinder, dropper, burette, spatula, reagent bottle, test tube stand, pipette stand, tripod stand, Bunsen burner, wire gauze, crucible, funnel, centrifuge tubes 2. Instruments: Separatory funnel, centrifuge, pH meter, Electric balance, hot plate	20
II	3. Determination of pH of various solutions using a pH meter – NaOH, sulphuric acid, distilled water 4. Preparation of Normal solution- NaOH 5. Preparation of percentage/ vol-vol solutions- Sulphuric acid 6. Preparation of buffers 7. Paper Chromatography- Isolation of the pigments from leaves of Radish 8. Qualitative analysis of Carbohydrate & fats	26
III	9. Preparation of stains. 10. Visualization of animal and plant cell by dye (methylene blue). 11. Study of polyploidy in Onion root tip by colchicine treatment.	24
IV	12. Study of different stages of Mitosis.	20

	13. Study of different stages of Meiosis. 14. Examination of blood samples: Blood grouping	
	Total	90

Textbooks:

1. An Introduction to Practical Biochemistry – 3rd edition, 2017, David Plummer.
2. Introductory Practical Biochemistry – 4th edition, 2017, S.K.Sawhney and RandhirSingh.

Reference books:

1. Practical Clinical Biochemistry, Harold Varley, 6th edition, 2022, CBS Publishers.
2. Practical Clinical Biochemistry: Methods and Interpretation, 4th edn. RanjnaChawla, Jaypee Brothers Medical Publishers.
3. Practical and Clinical Biochemistry for Medical Students, ed. T.N. Pattabhiraman, Gajanna Publishers.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

MINOR PAPER: Nutritional Biochemistry

Course level: 100

Subject Code:

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

The course aims at providing deep understanding of various nutrients, their metabolism, functions and deficiency diseases to students.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Understand the basic concepts of nutritional biochemistry, biochemical basis and nutritional importance of macronutrients, biochemical mechanisms for the symptoms of vitamin deficiencies and excesses, and mineral macronutrients.	BT1&BT2
CO 2	Apply the knowledge on their diet management, selection of nutrient-rich foodstuffs, maintaining good health, and distinguishing various diseases and disorders caused by nutrient deficiency	BT3 & BT4
CO 3	analyze the difference between macronutrients and micronutrients	BT4
CO 4	Analyze the application of Nutritional Biochemistry in human health and related research.	BT2 and BT3

Course Contents

Modules	Course content	Periods
I	Introduction to Nutrition and Energy Metabolism: Defining nutrition, role of nutrients. Unit of energy, biological oxidation of foodstuff. Physiological energy value of foods, SDA. BMR and RMR- factors affecting BMR. Recommended Nutrient Intakes (RNI) and Recommended Dietary Allowances for different age groups.	15
II	Macronutrients- Food sources of carbohydrates, Review functions of carbohydrates. Essential Fatty Acids; Functions of EFA, RDA, – excess and deficiency of EFA. Dietary implications of fats and oils, MUFA, PUFA and SFA. Importance of the following: a) Omega – fatty acids. Omega 3/omega 6 ratio b) Phospholipids c) Cholesterol in the body d) Mono, Polyunsaturated and Saturated Fatty Acids. Essential and Non-essential amino acids. Digestion of carbohydrates and proteins.	15
III	Micronutrients: Vitamin A, D, E, K and dietary sources, RDA, Deficiency. Role of Vitamin A as an antioxidant, and immunity. Role of Vitamin E as an	15

	antioxidant. Extra-skeletal role of Vitamin D and its effect on bone physiology .The B Complex vitamins- Dietary sources, RDA, role in metabolism and deficiency disease. Biochemical basis for deficiency symptoms.	
IV	Micro Minerals and Trace elements: Calcium, Iron and Phosphorus- Distribution in the body. Sources, RDA. Iodine, Fluoride, Mg,Cu,Zn,Se,Manganese,Chromium,Molybdenum.Distribution in the human body,Physiology, Function, deficiency, and Source.Antinutrients- source, role in health.	15
	Total	60

Textbooks:

1. Biochemistry by U. Satyanarayana & U. Chakrapani (6th edn), Elsevier, ISBN 978-8131264355, 2021
2. Textbook of Biochemistry for medical students by DM Vasudevan, Sreekumari S, and VK Vaidyanathan (8th edn), Jaypee Brothers Medical Publishers, ISBN- 978-9354656484, 2022

Reference books:

1. Coombs Jr. G. F., (2008). The vitamins, Fundamental aspects in Nutrition and Health.
2. Voet, D. and Voet, J.G. (2011) Biochemistry. Fourth Edition, John Wiley & Sons
3. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M. 7th Edition, 2017, WH Freeman and Company, New York, USA

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Group discussion Flipped classroom

SECPAPER: Biochemical assessment of food products

Course level: 100

Subject Code:

L-T-P-C: 0-0-6-3

Scheme of evaluation: Practical (P)

Course Objective

The aim of the course is to familiarize students with basic food components and their assessment in day to day used food products. The course will acquaint students with the methods of estimating the various basic food components like carbohydrates, lipids, proteins and macro/micro nutrients using different methods.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Outline and discuss the role of each biomolecule and their various food sources	BT1&BT2
CO 2	Review and demonstrate the different estimating methods for each biomolecule in the food sources.	BT2 & BT3
CO 3	Apply the knowledge of biomolecules practically and interpret experimental results which will be helpful in development of practical skills in biochemistry.	BT3
CO 4	Categorize and test the various components present in different food sources.	BT4

Course Contents

Modules	Course content	Periods
I	Carbohydrates: Introduction to carbohydrates. Determining the presence of carbohydrates in food samples like rice, fruit juice by Fehling's tests, Benedict's tests. To understand the color tests for reducing and non-reducing sugars. To test the presence of starch in food material (potato tubers, green leaves) and to reveal its biological importance and chemical nature.	24
II	Lipids: Classification of lipids. Role of lipid in human health. Lipid deficiency in human body. Determining the presence of fats and oils in almond seeds, groundnut etc. To test the presence of cellulose in the cotton threads or filter paper and to reveal the biological significance and chemical nature of cellulose.	24
III	Protein: Types of protein. Significance of protein in human health. Deficiency of protein and diseases related to it. Determining the presence of protein in food samples like pulses, egg, milk by Xanthoproteic Test, Biuret test, Millon's Test.	24
IV	Micro and macronutrients: Role and food sources of vitamins, macrominerals, microminerals and trace elements. Assessment of micronutrients (Ca, P, Fe, Vitamin A, and Vitamin C) status in food samples like milk, fruit juice, vegetables.	18
	Total	90

Textbooks:

1. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, USA, 7th edition, 2017
2. Victor Rodwell, David Bender, P. Anthony Weil, Peter Kennelly. Harpers Illustrated Biochemistry 31th Edition, 2018.

Reference books:

1. Practical Biochemistry – 3rd edition, 2017, David Plummer.
2. Introductory Practical Biochemistry – 4th edition, 2017, S.K.Sawhney and Randhir Singh.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

SYLLABUS 2nd SEMESTER

MAJOR PAPER: Proteins and Enzymes

Course level: 200

Subject Code: BCH152M201

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

To provide overview of protein biochemistry and enzymology to undergraduate students since proteins and enzymes are the most versatile functional entities in life. The biochemical, structural, functional and aspects of interaction of proteins and enzymes will be introduced in this course.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Describe and summarize the molecular, chemical and structural	BT1 & BT 2

	foundations of proteins and enzymes.	
CO 2	Discuss and interpret core concepts of enzyme kinetics and activity	BT2 & BT3
CO 3	Apply their knowledge in various applications associated with life sciences research as well as in industry and biomedicine.	BT 3
CO 4	Analyze outcomes of performed laboratory experiments related to protein biochemistry and enzymology.	BT 4

Course Contents

Modules	Course content	Periods
I	Introduction to proteins and their structural organization: Amino acids and their properties. Peptides and their biological significance - hormones, antibiotics and growth factors. Diversity of proteins and their functions. Protein sequence - Edman degradation. Solid phase peptide synthesis. Organization of protein structure -primary, secondary, tertiary and quaternary structures. Conjugated proteins, multimeric proteins and metalloproteins. Bonds in protein structures-covalent and non-covalent. Dihedral angles. Ramachandran map, Secondary structure - helices, sheets and turns.	15
II	Three-dimensional structures and protein folding Characteristics of tertiary and quaternary structures. Motifs and domains. Structure-function relationship in proteins. 3D structures of myoglobin and hemoglobin. Oxygen binding curves, influence of pH and effector molecules. Concerted and sequential models for allosteric proteins. Hemoglobin disorders. Protein folding-denaturation and renaturation. Role of chaperones. Protein misfolding and aggregation diseases.	15
III	Introduction to enzymes, their characteristics and kinetics Nature of enzymes - protein and non-protein (ribozyme, abzymes). Cofactor and prosthetic group, apo- and holo-enzymes. Features of enzyme catalysis. Classification of enzymes and nomenclature. Fischer's lock& key and Koshland's induced fit hypothesis. Enzyme specificity. Enzyme kinetics- Michaelis-Menten equation, Lineweaver-Burk plot. Determination of K_m , V_{max} , K_{cat} . Factors affecting enzyme activity. Enzyme Inhibition: Reversible (competitive, uncompetitive, non-competitive) and irreversible inhibition. Mechanism based inhibitors.	15
IV	Mechanism of enzyme action and enzyme regulation General mechanisms of action. Acid-base and covalent catalysis	15

	(chymotrypsin, lysozyme). Metal activated enzymes and metalloenzymes. Allosteric regulation and feedback inhibition (ATCase). Reversible covalent modification (glycogen phosphorylase). Proteolytic cleavage zymogen. Multienzyme complex. Coenzymes.	
	Total	60

Textbooks:

1. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, USA. 7th edition, 2017.
2. Lodish, H., Berk, A., Zipursky, S. L., Matsudaira, P., Baltimore, D. and James Darnell, J. Molecular Cell Biology, Freeman, 9th edition 2021.

Reference books:

1. Berg, J. M., Tymoczko, J. L. and Stryer., Biochemistry,, W.H Freemanand Co., 9th edition, 2019.
2. Alberts, B., Johnson, A., Lewis, J., Raff, M., Roberts, K. and Walter P. Molecular Biology of the Cell. Garland Science, 7th edition, 2022.
3. Branden, C. &Tooze, J. Introduction to Protein Structure. Garland, 2nd edition,1999.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom Group discussion
MAJORPAPER: Proteins and Enzymes Practical Course level: 200 Subject Code:BCH152M212 L-T-P-C: 0-0-6-3		

Scheme of evaluation: Practical (P)

Course Objective

Objective of the course is to get students exposed to a biochemistry laboratory and develop good laboratory practices with hands on experience of simple but important biochemical experiments.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Describe the working principles of various qualitative and quantitative analysis methods.	BT 2
CO 2	Employ a hands-on experience on handling various laboratory equipments, chemicals and instruments.	BT 3
CO 3	Apply and translate the knowledge in basic research projects, biomedical research, clinical applications, industrial applications, etc.	BT2 & BT 3
CO 4	Analyze and criticize the working principles, significance of practical results	BT 4

Course Contents

Modules	Course content	Periods
I	1. Qualitative tests for amino acids 2. Qualitative tests for proteins 3. Estimation of proteins using UV absorbance and Biuret method. 4. Estimation of proteins using Lowry/Bradford method.	24
II	5. Assay of salivary amylase 6. Effect of pH on enzyme activity 7. Effect of temperature on enzyme activity. 8. Determination of K_m and V_{max} using Lineweaver-Burk plot	24

III	9. SDS-PAGE analysis of protein. 10. Separation of amino acids by paper chromatography	20
IV	11. Separation of photosynthetic pigments by TLC. 12. Isolation of chloroplast from spinach leaves, estimation of chlorophyll and photosynthetic activity.	22
	Total	90

Textbooks:

1. Practical Biochemistry – 3rd Edn. David Plummer.
2. Introductory Practical Biochemistry – S. K. Sawhney and Randhir Singh.

Reference books:

1. Practical Clinical Biochemistry, ed. Harold Varley, 4th edn. CBS Publishers.
2. Practical Clinical Biochemistry: Methods and Interpretation, 4th edn. Ranjna Chawla, Jaypee Brothers Medical Publishers.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

MINOR PAPER: Introduction to Cytology

Course level: 100

Subject Code: BCH152N201

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

The objective of this paper is to offer insights into the basic structure and function of a cell and cellular organelles as well as on the various techniques. Students will understand the structures and purposes of basic components of prokaryotic and eukaryotic cells, especially macromolecules, membranes, and organelles.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Recall the types of cells and distinguish their cellular components	BT2 & BT3
CO 2	Recognize the role of different cell organelle structure in relation to cell physiology	BT3
CO 3	Summarize various cell biology processes such as cell cycle, cell division, types of cell death and vesicular transport and apply this knowledge to predict deviations from normal cell physiology	BT4
CO 4	Analyze the application of cell biology in research.	BT2 and BT3

Course Contents

Modules	Course content	Periods
I	Introduction to cell structure: Introduction to cell structure: Prokaryotic (archaea and eubacteria) and eukaryotic cell (animal and plant cells). Structure of different cell organelles: nuclear envelope and membrane, ER, Organization of Golgi apparatus, Ribosomes, Lysosome. Structure and functions of mitochondria, chloroplasts and peroxisomes. Biogenesis of Mitochondria and chloroplast. Cell recognition and membrane transport.	15
II	Cell wall and extracellular matrix: Prokaryotic and eukaryotic cell wall, cell-matrix proteins. Types of Cytoskeletal proteins: Tubulin, Actin and Intermediate filaments. Cell-matrix interactions and cell-cell interactions. Adherence junctions, tight junctions, gap junctions, desmosomes, hemidesmosomes, focal adhesions, and	15

	plasmodesmata. Cilia and Flagella.	
III	Cell cycle, cell division, and cell death: Eukaryotic cell cycle, cell cycle stages: G ₀ , G ₁ , S, G ₂ , M phase, restriction point, and checkpoints. Chromosomal organization, Sister chromatids and nucleosomes. Cell division: concepts of mitosis and meiosis.	15
IV	Vesicular transport and membrane fusion: Types of vesicle transport and their function. Molecular mechanism of vesicular transport. Phagocytosis, Pinocytosis and receptor mediated endocytosis: General Mechanism.	15
	Total	60

Textbooks:

1. Molecular Biology of the Cell, 7th edition, 2022, Bruce Alberts, Dennis Bray, Julian Lewis, Martin Raff, Keith Roberts, and James D Watson. Publisher New York: Garland Science
2. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, USA. 7th edition, 2017.

Reference books:

1. Lipid Biochemistry; 5th Edn. Michael I. Gurr, John L. Harwood and Keith N. Frayn, Blackwell Science (2002).
2. Biochemistry Ed. Donald Voet & Judith G. Voet, 4th Edn. John Wiley & Sons, Inc. (2012).

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom Quizzes

SEC PAPER: Food Adulteration

Course level: 200 Subject Code: BCH152S211 L-T-P-C: 0-0-6-3 Scheme of evaluation: Practical (P)

Course Objective

To educate about common food adulterants and their effects on health as well as to teach them various biochemical analysis methods for detection of food adulterants.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Identify classes of foods and their contamination by various adulterants	BT 2
CO 2	Describe rules and regulations in relation to the control of food adulteration and power and activity of FSSAI.	BT 3
CO 3	Examine various biochemical methods and analyze adulterants in milk and milk products, oils and fats, and salt and spices, confectionary, food grains, and beverages etc.	BT2 & BT 3
CO 4	Analyze and criticize the working principles, significance of practical results	BT 4

Course Contents

Modules	Course content	Periods
I	1. Adulteration – definition, types; New adulterants in foods. 2. Physical test for adulteration of milk - Physical Tests, Detergent Test, Flow test 3. Chemical test for adulteration of milk - Test for starch in Milk, Test for cane sugar in Milk, Test for soda in milk	24

II	4. Adulteration of ghee – test for vegetable fat – nitric acid test, soda ash test 5. Test for added alkali - Baudovin test 6. Adulteration of Paneer: Presence of starch in paneer 7. Test for rancidity in oils	22
III	8. Coriander powder: Test for starch & horse dung powder 9. Chilli powder. Test for oil soluble dyes, powdered bran, saw dust and brick powder 10. Turmeric Powder: Test for metanil yellow and lead chromate polish	22
IV	11. Wheat flour: Test for sand, dirt, husk and chalk powder 12. Whole grain; adulteration with datura 13. Test for urea in parched rice	22
	Total	90

Textbooks:

1. A laboratory manual on Food Adulteration, Shalini Sehgal, Wiley Publications, 2020, ISBN- 9389633230
2. Food adulteration and its detection, Jesse Park, Wentworth Press, ISBN-9781362422099

Reference books:

1. Food Science, B Srilaxmi, 7th Edition, New Age Publication, 2018, ISBN- 9386418894
2. <https://fssai.gov.in/cms/food-safety-and-standards-act-2006.php>

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

SYLLABUS 3rd SEMESTER

MAJOR (Core) PAPER: Metabolism of Biomolecules

Course level: 200

Subject Code: BCH152M301

L-T-P-C: 3-1-0-4

Scheme of evaluation: Theory (T)

Course Objective

The objective of this course is to provide an understanding of metabolism of all the biomolecules, the enzymes involved in such metabolic pathways, their regulation, and importance with regards to metabolic disturbances.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Understand the basic concept of metabolism: catabolism and anabolism.	BT1 & BT 2
CO 2	Remember the various metabolism of biomolecules: biosynthesis and	BT1

	degradation pathways and their regulations.	
CO 3	Apply the knowledge of metabolic pathways to understand how defects in concentration and imbalanced functioning of the participating substrate and enzymes may lead to several disorders.	BT 3
CO 4	Analyze the various disorder's observations and plan targeted therapeutic approaches to tackle metabolic diseases, especially those related to lipid, protein, nucleic acid, and carbohydrate metabolism.	BT 4

Course Contents

Modules	Course content	Periods
I	Concept of metabolism, catabolism, and anabolism, Carbohydrate metabolism : glycolysis, gluconeogenesis, fates of pyruvate, formation of acetyl coenzyme A, TCA cycle, ATP yield from complete oxidation of glucose, pentose phosphate pathway. Oxidative Phosphorylation and Electron Transport Chain.	15
II	Basic structure of Phospholipids, saturated and unsaturated fatty acids, Beta-oxidation of fatty acids, biosynthesis of saturated and unsaturated fatty acids (precursors and site of synthesis, ketone bodies).	15
III	Amino acid metabolism General reactions of amino acid metabolism, (oxidative deamination, transamination, decarboxylation etc.), glucogenic and ketogenic amino acids, urea cycle, biosynthesis of essential and non-essential amino acids.	15
IV	Biosynthesis and catabolism of purines and pyrimidine, Preliminary idea of De novo synthesis and Salvage pathway, Regulation and disease due to defect in nucleotide metabolism.	15
	Total	60

Textbooks:

1. Principles of Biochemistry, 8th Edition. Nelson, D.L. and M.M. Cox (2021). WH Freeman and Co. ISBN: 9781319381493.
2. Fundamentals of Biochemistry: Life at the molecular level, 4th Edition. D. Voet, J.G. Voet and W. Pratt (2016). John Wiley & Sons Inc. ISBN: 978-1-118-91840-1.

Reference books:

1. Principles of Biochemistry, 4th Edition. Robert Horton H, Laurence A Moran, Gray Scrimgeour K (2006). Pearsarson Publisher. ISBN-13: 978-0321707338.
2. Rodwell, V. W., Bender, D., & Botham, K. M. (2018). *Harper's illustrated biochemistry*. McGraw-Hill.
3. Introductory Practical Biochemistry. SK Sawney and R. Singh (2000). Narosa Publisher. ISBN 9788173193026
4. An Introduction to Practical Biochemistry 3rd Edition. Plummer D.T. (1998). Tata McGrawhill Publication. ISBN: 1118357728.
4. Satyanarayana, U. (2013). *Biochemistry*. Elsevier Health Sciences.
5. Michal, G., & Schomburg, D. (Eds.). (2012). *Biochemical pathways: an atlas of biochemistry and molecular biology*. John Wiley & Sons.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom

MAJOR (Core) PAPER: Metabolism practical

Subject Code: BCH152M312

L-T-P-C: 0-0-8-4

Scheme of evaluation: Practical (P)

Course Objective

Objective of the course is to make students understand methods of assessing levels of different physiologically important molecules such as glucose, pyruvate, specific amino acids, triacylglycerol, and cholesterol and then compare the level with normal physiological level.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Classify the various biological samples and hands-on training to understand the basic working techniques.	BT 2
CO 2	Understand how glucose, pyruvate, proteins, cholesterol, and neutral triacylglycerol levels can be measured in various samples, such as blood and urine.	BT 1
CO 3	Apply the gained knowledge to analyze and understand how changes in the physiological level of glucose, triacylglycerols, cholesterol, etc., ultimately result in various health issues.	BT 3& BT 4
CO 4	Apply a preliminary basic experimental approach to understanding the level of neutral lipid content, glucose, cholesterol, etc., normal vs. disease models.	BT3 & BT4

Course Contents

Modules	Course content	Periods
I	1. Colorimetric estimation of soluble sugars 2. Colorimetric estimation of pyruvate	24

	3. Measurement of soluble sugars and pyruvate in yeast/bacterial cells in different growth conditions	
II	4. Colorimetric estimation of amino acids 5. Colorimetric estimation of proteins (Biuret method) 6. Estimation of tryptophan residues in a protein.	24
III	7. Measurement of respiration by oxygen consumption. 8. Effects of inhibitors / uncouplers on respiration 9. Measurement of NAD/NADH in yeast/bacterial cells.	24
IV	10. Estimation of blood glucose. 11. Isolation and estimation of triacylglycerols. 12. Isolation of cholesterol from egg yolk and its estimation.	24
	Total	96

Textbooks:

1. Practical Biochemistry – 3rdEdn. David Plummer.
2. Introductory Practical Biochemistry – S.K.Sawhney and Randhir Singh.

Reference books:

1. Practical Clinical Biochemistry, ed. Harold Varley, 4th ed. CBS Publishers.
2. Practical Clinical Biochemistry: Methods and Interpretation, 4th ed. Ranjna Chawla, Jaypee Brothers Medical Publishers.
3. Introductory Practical Biochemistry. SK Sawney and R. Singh (2000). Narosa Publisher. ISBN 9788173193026
4. An Introduction to Practical Biochemistry 3rd Edition. Plummer D.T. (1998). Tata McGrawhill Publication. ISBN: 1118357728.
4. Tiwari, A. (2015). *Practical biochemistry: A student companion*. LAP Lambert Academic Publishing.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

MINOR PAPER: CLINICAL BIOCHEMISTRY

Subject Code: BCH152N301

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

To introduce the various parameters that determine a healthy and diseased state and to understand the workings of disease progression and development.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Identify the method of specimen collection and analysis and outline metabolic disorders, syndromes arising out metabolic disorders and the biochemistry of cancer.	BT 1
CO 2	Identify and review the various aspects clinical biochemistry, their implications in health, causes and implications of metabolic disorders and the biochemical basis of cancer.	BT 1 and BT 2
CO 3	Apply the knowledge gained to deliberate on clinical findings, check the relation of metabolites and syndromes associated with build-up of metabolites and the biochemical reasons leading to cancer progression.	BT 3
CO 4	Analyze the relation of body metabolites and their deregulation with appearance of metabolic syndromes and relate the build-up of cancer specific conditions in the progression of cancer.	BT4

Course Contents

Modules	Course content	Periods
I	Specimen collection and analysis: Concepts of accuracy, Precision, Reliability, Reproducibility, Normal values, Specimen collection and processing, Blood	15

	Collection-Anticoagulants, Venipuncture, Urine collection, CSF, Amniotic fluid, pH of blood, acid base equilibrium, sodium, potassium, chloride, bicarbonate.	
II	Inborn errors of metabolism: Carbohydrate metabolism: galactosemia, Normal levels, renal threshold, Factors influencing blood glucose, Glycogen storage disorders, Pentosuria Lipid metabolism: Hyperlipidemia, Hyperlipoproteinemia, Fatty liver.	15
III	Amino acid metabolism-alkaptonuria, maple syrup urine disease, phenylketonuria, homocystinuria, proteinuria, albinism, multiple myeloma. Nucleotide metabolism: Lesch-Nyhan Syndrome, Biochemistry of anemia, thalassemia, porphyria	15
IV	Disease caused by metabolic disorders of endocrine glands: Diabetes, obesity, atherosclerosis, Conn's syndrome, Addison's disease, Cushing's syndrome, hypo- and hyperthyroidism, gonadal dysfunction, dwarfism and gigantism, Diseases caused by chromosomal abnormalities- Down, Turner and Klinefelter syndromes.	15
	Total	60

Textbooks:

1. Textbook of medical biochemistry by S. Ramakrishnan, K G Prasanna 3rd Edition.
2. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M. 7th Edition, 2017, WH Freeman and Company, New York, USA.

Reference books:

1. A C Deb Fundamentals of Biochemistry, 10th edition, (2018), New Central Book Agency, London.
2. Berg, J.M., Tymoczko, J.L. and Stryer. Biochemistry, W.H. Freeman and Co., 9th edition, 2019.
3. Allan Gaw, Michael Murphy, Rajeev Srivastava, Robert Cowan, Denis O'Reilly, Clinical Biochemistry, 6th Edition, 2017
4. Gaw, A., Murphy, M., Srivastava, R., Cowan, R. A., & O'Reilly, D. S. J. (2013). *Clinical Biochemistry E-Book: An Illustrated Colour Text*. Elsevier Health Science
5. Blanco, A., & Blanco, G. (2017). *Medical biochemistry*. Academic Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom

SEC PAPER: Working with proteins

Subject Code: BCH152S311

L-T-P-C: 0-0-4-2

SCHEME OF EVALUATION: Practical (P)

Course Objective

To introduce the students to the basic laboratory skills needed to work with proteins.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Define the biochemistry behind protein isolation	BT 1
CO 2	Explain the differences in isolating proteins from different sources.	BT 2
CO 3	Analyze the importance of transferring proteins from gel to the membrane	BT4
CO 4	Compare the different amino acid functional groups in the protein.	BT2

Course Contents

Modules	Course content	Periods
I	Isolation of proteins from – 1. Blood 2. Plant tissue 3. Animal tissue	6
II	Estimation and separation of protein- 1. Estimation of isolated protein using Bradford, Lowry methods 2. Separation of proteins using chromatographic and electrophoresis methods	6
III	Transferring proteins to membrane 1. Demonstration of transferring proteins to membrane from gel and detection using Indian Ink staining 2. Demonstration of immunoblotting of specific proteins.	6
IV	Amino acid estimation in protein 1. Estimation of thiol group in a protein. 2. Estimation of tryptophan residues in a protein	6
	Total	24

Textbooks:

1. Plumer D.T. (2017) An Introduction to Practicals in Biochemistry, Tata McGraw-Hill.
2. Jayaraman J. (2011) Laboratory Manual in Biochemistry, New Age Int. Pub.

Reference books:

1. Sadasivam S. and Manickam A. (2018) Biochemical Methods, New Age Int. Pub. Delhi.
2. Segal I.H (2010) Biochemical Calculations, John Wiley.
3. Gupta P.P and Gupta N. (2018) Essentials of Practical Biochemistry, Jaypee Brothers med. pub.
4. Introductory Practical Biochemistry. SK Sawney and R. Singh (2000). Narosa Publisher. ISBN 9788173193026 4. An Introduction to Practical Biochemistry 3rd Edition. Plummer D.T. (1998). Tata McGrawhillPublication. ISBN: 1118357728.
5. Practical Clinical Biochemistry: Methods and Interpretation, 4th ed. Ranjna Chawla, Jaypee Brothers Medical Publishers.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

SYLLABUS 4th SEMESTER

MAJOR (Core) PAPER: Concepts of Genetics

Course level: 200

Subject Code: BCH152M401

L-T-P-C: 3-1-0-4

SCHEME OF EVALUATION: Theory (T)

Course Objective

The aim of the course is to provide knowledge of both mendelian and non-mendelian concepts in genetics to students with a focus on the molecular basis of genetics.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Remember various topics under Mendel's Principle of heredity, gene function and genetics in bacteria, Pedigree analysis, Linkage, crossing over, mapping techniques, and chromosomal aberrations.	BT1 & BT 2
CO 2	Understand Mendel's laws and ratios; relationship between genetic inheritance, mechanisms of genetic exchange in prokaryotes; the of concept of recombination and linked genes; use recombination frequencies to determine gene order and distance; genetic mapping in eukaryotes using test crosses; the difference in the genetic basis of sex determination in Humans and Drosophila.	BT 2& BT 3
CO 3	Apply their knowledge in testing genetic hypothesis through statistical tools; finding recombination frequencies to determine gene order and	BT 3

	distance; to build genetic mapping in eukaryotes.	
CO 4	Analyze inheritance pattern, cytogenetics mapping, pedigree analysis and chromosomal aberrations.	BT 4

Course Contents

Modules	Course content	Periods
I	Mendel Principles of Heredity- Basic principles of heredity, Mendel's laws of segregation and independent assortment, test cross, expression and interaction of genes, complimentary genes and epistasis. Sutton and Boveri hypothesis. Multiple alleles and pleiotropism	15
II	Linkage and crossing over- Basic principles of linkage and crossing over, Morgan' experiments showing linkage. Crossing over and genetic recombination. Examples of crossing over and genetic recombination in drosophila. Linkage maps of drosophila chromosomes and human chromosomes.	15
III	Bacterial genetics- Analysis of bacterial genetics, auxotrophs, prototrophs. Discovery of conjugation in bacteria, sex factor, transfer of sex factor, Hfr strains. Transformation, natural transformation and engineered transformation. Transductiongeneralized transduction and specialized transduction. Application of conjugation, transformation and transduction in mapping of genes.	15
IV	Basics of human genetics- Pedigree analysis, chromosomal numerical abnormalities- Trisomy and monosomy. Euchromatin and heterochromatin. Sex chromosome, sex-linked inheritance, sex determination, dosage compensation and chromosomal aberrations	15
	Total	60

Textbooks:

1. Genetics,Snustad, D.P. and Simmons, M.J., John Wiley & Sons. (Singapore),ISBN: 978-1-118-09242-2., 7th edition, 2015
2. Genetics - A Conceptual Approach,Pierce, B.A., W.H. Freeman & Co. (NewYork), ISBN:13:978-1-4292-7606-1 / ISBN:10:1-4292-7606-1., 4th edition,2012

Reference books:

1. iGenetics, P.J.Russell,Benjamin/Cummings,Pearson Education 3rd edition, ISBN- 978-0321569769, 2009
2. Principles of Genetics, DP Snustad and MJ Simmons, John Wiley & Sons Inc, ISBN: 978-1-119-14228-7, 2015
3. Concepts of Genetics, Williams S Klug, Michael R Cummings, Charolette Spencer and Michael APalladino
4. Genetics, P.K. Gupta, Rastogi Publication • Concepts of Genetics (Sixth Edition), William S. Klug and Michael R, Cummings, Pearson Education
4. Jorde, L. B., Carey, J. C., & Bamshad, M. J. (2015). *Medical genetics e-Book*. Elsevier Health Sciences.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom

MAJOR (Core) PAPER: Human diseases and Basic Ayurveda

Subject Code: BCH152M402

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

To introduce students about the major diseases in India and the basics of ayurveda.

Course Outcome

On Successful Completion of the course, the students will be able to:

Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Define the multifactorial causes of diseases, facilitating better prevention and treatment strategies	BT 1
CO 2	Explain the interrelation of diseases	BT 2
CO 3	DefineAyurvedic philosophy, diagnosis, and treatment modalities	BT 1 and BT 2
CO 4	Analyze the role of medicinal herbs in ayurveda	BT 4

Course Contents

Modules	Course content	Periods
I	Etiology of Diseases: Introduction to Disease Etiology; Genetic Factors in Disease; Environmental Factors;Lifestyle Factors; Immunological Factors. Overview of the global burden andepidemiology of diabetes, obesity, and atherosclerosis. Pathophysiology of Diabetes: Mechanisms underlying insulin resistance and beta-celldysfunction leading to complications of diabetes. Pathogenesis of Obesity: Discussion on the role of genetic, environmental, and behavioralfactors in obesity development. Atherosclerosis and Cardiovascular Disease: Pathogenesis of atherosclerosis, and itsrelationship with heart attack and stroke.	15
II	Interrelation of Diseases: Concept of disease comorbidity, common pathways and mechanisms; syndromicassociations, secondary diseases arising from primary conditions, side effects ofmedications, disease clustering in populations. Understanding Cancer: Overview of cancer types and prevalence; Introduction to cancergenetics and risk factors; Cell cycle dysregulation and genetic mutations; Major cancetypes and their characteristics; Understanding tumor heterogeneity; Imaging	15

	and molecular diagnostic methods; Importance of early detection and screening; Overview of surgery, radiation, and chemotherapy; Targeted therapies and immunotherapy approaches.	
III	Basic concepts of Ayurveda: Origin and history of Ayurveda; Ayurvedic concepts of health and disease. Concept of Panchamahabhutas (Five Elements: space, air, fire, water, and earth and their relation with the human body); Tridosha Theory (Vata, Pitta, and Kapha); SaptaDhatu (Seven Tissues); Agni (Digestive Fire); Ama (Toxins); Ayurvedic Diagnosis (Techniques of Ayurvedic diagnosis); Ayurvedic Lifestyle Practices (Daily routine, Dietary guidelines, Importance of yoga, meditation, and pranayama for health); Ayurvedic Therapies (Detoxification therapies, remedies, oils, and massages).	15
IV	Role of Herbal Medicine in Ayurveda:- History and significance of medicinal herbs in Ayurvedic tradition; Classification of ayurvedic herbs based on properties and actions. Therapeutic properties, traditional uses and health benefits of Amla (<i>Embolia officinalis</i>), Ashwagandha (<i>Withania somnifera</i>), Neem (<i>Azadirachta indica</i>), Triphala, Tulsi (<i>Ocimum sanctum</i>).	15
	Total	60

Textbooks:

1. Lehninger - Principles of Biochemistry; David L. Nelson and Michael M. Cox, 6th Edition, W. H. Freeman (2013). 46

Alagappan R (2017) Medicine For Ayush Students 1st Edition , Atithi books, New Delhi

Reference books:

1. Biochemistry- The Chemical Reactions of Living Cells; David E. Metzler, 2nd Edition, Academic Press(2001).
2. Outlines of Biochemistry; Eric E. Conn, Paul K. Stumpf, George Breuning, Roy H. Doi, 5th Edition, John-Wiley and sons(2009).
3. Biochemistry- The Chemical Reactions of Living Cells; David E. Metzler, 2nd Edition, Academic Press (2001).
4. Ravindra Angadi (2017) Classical Compendium Of Ayurvedic Pharmaceutical Science, Athithi Medical books Pvt Ltd, New Delhi.
5. Kshirsagar, M., & Magno, A. C. (2012). *Ayurveda: a quick reference handbook*. Lotus

Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Quiz Seminar Flipped classroom

MAJOR (Core) PAPER: Genetics Practical

Subject Code: BCH152M412

L-T-P-C: 0-0-8-4

SCHEME OF EVALUATION: Practical (P)

Course Objective

The objective of the course to introduce the students the basic knowledge of simple experiment related to the field of genetics such as pedigree analysis, karyotyping in diseased and normal conditions., isolation of plasmid DNA, restriction-digestion of plasmid DNA, and amplification of a DNA of interest towards understanding of basics of genetic engineering.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Understand and have a practical knowledge of karyotyping of human	BT 1& BT 2

	chromosomes, pedigree analysis, plasmid isolation and restriction digestion of the same, and amplification of a DNA of interest.	
CO 2	Apply the knowledge gained and the basic techniques learnt in setting up research question in future involving in human genetic disorders and basic techniques to be undertaken while tackling them through genetic engineering	BT 3
CO 3	Analyze the online databases such as NCBI Nucleotide, primer designing sites which would help them in future application related to DNA.	BT 3 & BT 4

Course Contents

Modules	Course content	Periods
I	1. Buccal smear for Barr body analysis 2. Karyotyping of normal and turner/down's syndrome from supplied photographs	24
II	3. Isolation and analysis of plasmid DNA from bacteria 4. Restriction digestion of Plasmid DNA	24
III	5. Preparation of chromosome from mouse bone marrow 6. Study of different stages of meiosis in grasshopper testis	24
IV	7. Designing of primer for PCR 8. Amplification of DNA using PCR 9. Agarose gel electrophoresis of PCR product	24
	Total	96

Textbooks:

1. Brown, T. A. Gene Cloning and DNA analysis: an introduction. Blackwell Science, 6th edition 2010.
2. Genetics, Snustad, D.P. and Simmons, M.J., John Wiley & Sons. (Singapore), ISBN: 978-1-118-09242-2., 7th edition, 2015

Reference books:

1. Modern Experimental Biochemistry. 3rd Edition. Rodney Boyer
2. Advanced Methods in Molecular Biology and Biotechnology- A Practical Lab Manual. Authors: Khalid Z Masoodi, Sameena Maqbool Lone, and Rovidha Saba Rasool
3. Principles and Techniques of Biochemistry and in Molecular Biology. Keith Wilson and

John Walker

4. Ptashne, M and Gann, A. Genes and Signals, Cold Spring Harbor Laboratory Press. 2001
5. Watson, J. D. et al. Molecular Biology of the Gene. Benjamin Cummings, 7th edition 2013.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

MINOR PAPER: General Microbiology

Subject Code: BCH152N401

L-T-P-C: 2-1-0-3

SCHEME OF EVALUATION: Theory (T)

Course Objective

The course will provide the fundamental concepts of Microbiology including structure and classification of bacteria, microbial diversity and pathogenesis.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level

CO 1	Define the different types of microorganisms, their importance, and pathogenesis	BT 1
CO 2	Compare the microbial diversity and physiology	BT 2
CO 3	Apply their knowledge in exploring microbial growth under various situations.	BT 3
CO 4	Analyze the pathogenic characteristics of different microbes	BT 4

Course Contents

Modul es	Course content	Periods
I	History of Microbiology - History of microbiology as a discipline, Spontaneous generation versus biogenesis, contributions of Anton von Leeuwenhoek, Joseph Lister, Paul Ehrlich, Richard Petri, Charles Chamberland, Edward Jenner, Louis Pasteur, Robert Koch, Martinus W. Beijerinck, Sergei Winogradsky, Alexander Fleming, Elie Metchnikoff and Emil vonBehring	15
II	Microbial Cell organization and Diversity of Microbial world Difference between prokaryotic and eukaryotic microorganisms. General characteristics of different groups: Acellular microorganisms (Viruses, Viroids, Prions) and Cellular microorganisms (Bacteria, Archaea, Algae, Fungi and Protozoa) with emphasis on distribution, occurrence and morphology. Cell-wall: Composition and detailed structure of Gram positive and Gram-negative cell walls, mechanism of Gram's staining. Cell Membrane: Structure, function and chemical composition of bacterial and archaeal cell membranes.	15
III	Microbial Nutrition and Growth Nutritional types of microorganisms, growth factors, culture media- synthetic and complex, types of media; isolation of pure cultures, growth curves, mean growth rate constant, generation time; influence of environmental factors on growth of microbes: effect of pH, temperature, solute, oxygen concentration, pressure and radiations. Sterilization, disinfection and antiseptics. Use of physical methods (heat, low temperature, filtration, radiation) and chemical agents (phenolics, halogens, heavy metals, sterilizing gases) in microbial control .	15
IV	Pathogenicity of Microorganisms and Antimicrobial Chemotherapy- Introduction to pathogenic microbes; Bacteria, Viruses, Algae, protozoa and fungi. General Characteristics of antimicrobial drugs, determining the level of microbial activity: dilution susceptibility test and disc diffusion test.	15

	Range of activity and mechanism of action of penicillin, vancomycin and tetracycline.	
	Total	60

Textbooks:

1. Lippincott's Illustrated Review of Microbiology by William A Strohl, 5th edition, 2017
2. Willey, J., Sherwood, L., Woolverton, C. (2017). Prescott's Microbiology (10th ed.). McGraw Hill international. ISBN 13:9781259657573

Reference books:

1. Chan, M.J., Krieg E. C. S., Pelczar, N. R. (2004) Microbiology (5th ed.). McGraw Hill International. ISBN 13:9780094623206.
2. Prescott, Harley, Wiley, J.M., Sherwood, L.M., Woolverton, C.J. (2018). Klein's Microbiology. (7th ed.). McGraw Hill International Edition (New York) ISBN: 978- 007-126727
3. Lehninger - Principles of Biochemistry; David L. Nelson and Michael M. Cox, 6th Edition, W. H. Freeman (2013).
5. iGenetics, P.J. Russell, Benjamin/Cummings, Pearson Education 3rd edition, ISBN- 978-0321569769, 2009 2. Principles of Genetics, DP Snustad and MJ Simmons, John Wiley & Sons Inc, ISBN: 978-1-119-14228-7, 2015

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MINOR PAPER: Biochemical Correlations of Diseases

Subject Code: BCH152N402

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

To provide knowledge about common lifestyle disorders and the molecular details of their development.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Review the basic relation between cells, tissues and organs	BT 2
CO 2	Define and describe the causes of different diseases	BT1 & BT2
CO 3	Discover the different underlying causes of diseases	BT3
CO 4	Explain the functioning of vaccines	BT 5

Course Contents

Modules	Course content	Periods
I	Introduction and brief description of cells, tissues and organs, their functions; Body fluids and their composition. Introduction to molecules as building blocks. Definition and differentiation of disease and disorder, types and causes.	15
II	Diabetes, major causes of diabetes, major genes involved in diabetes. Obesity, major genes involved in obesity. Cardiovascular diseases. Major underlying causes. Genes involved in CVD.	15

III	Alcoholic liver disease and fatty liver- Biochemical mechanisms, Causes, Prevention and dietary management. Nephrotic syndrome, Acute and Chronic renal failure- Biochemical mechanisms, Causes, Prevention and dietary management	15
IV	Viral infections- HIV/AIDS: Biochemistry of HIV infection, ART and social issues. Vaccines against HIV. Biochemistry of SARS-COVID I and II viral infections.	15
	Total	60

Textbooks:

1. Biochemistry; Donald Voet, Judith G. Voet, 4th Edition, John Wiley and sons (2010).
2. Lehninger - Principles of Biochemistry; David L. Nelson and Michael M. Cox, 6th Edition, W. H. Freeman (2013). 46

Reference books:

1. Biochemistry- The Chemical Reactions of Living Cells; David E. Metzler, 2nd Edition, Academic Press(2001).
2. Outlines of Biochemistry; Eric E. Conn, Paul K. Stumpf, George Breuning, Roy H. Doi, 5th Edition, John-Wiley and sons(2009).
3. Biochemistry- The Chemical Reactions of Living Cells; David E. Metzler, 2nd Edition, Academic Press (2001).
4. Gaw, A., Murphy, M., Srivastava, R., Cowan, R. A., & O'Reilly, D. S. J. (2013). *Clinical Biochemistry E-Book: An Illustrated Colour Text*. Elsevier Health Science
5. Blanco, A., & Blanco, G. (2017). *Medical biochemistry*. Academic Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

SYLLABUS 5th SEMESTER

MAJOR PAPER: Molecular Biology

Course level: 300

Subject Code: BCH152M501

L-T-P-C: 3-1-0-4

SCHEME OF EVALUATION: Theory (T)

Course Objective

The aim of the course is to provide knowledge of the central role of the information molecule – the DNA, in controlling all physiological and cellular processes.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Recall various topics on the flow of information from nucleic acids to proteins, and the significance of the central dogma	BT1 & BT 2
CO 2	Understand the basics of prokaryotic and eukaryotic molecular mechanisms and regulation of DNA replication, repair, transcription, post-transcriptional processing and protein synthesis.	BT 2 & BT 3
CO 3	Apply the knowledge of gene expression and regulation in their basic research projects, medical science, agriculture, industry, etc	BT 3
CO 4	Analyze the genetic differences and molecular mechanisms regulating gene expression between eukaryotes and prokaryotes.	BT 4

Course Contents

Modules	Course content	Periods
I	Introduction and History of Molecular Biology: Discovery of DNA as genetic material: experiments of Griffith, Avery, McLeod and McCarty, Hershey and Chase experiment, Lederberg and Tatum's conjugation experiment and Friedrich Miescher's experiment. Types of DNA and RNA. Chromosomal organizations in prokaryotes and eukaryotes. Chemical nature of the gene. Gene and gene concept: cistron, muton, replicon and recon. Central dogma of molecular biology.	12
II	DNA Replication and Repair Mechanism: Messelson–Stahl experiment of semiconservative DNA replication. DNA replication in prokaryotes and eukaryotes: enzymes and proteins involved in replication. DNA mutation and repair: types of mutation (mismatch, base-excision, nucleotide excision and direct repair). Replication of plasmids and mitochondrial DNA.	12
III	Transcription and Translation: Transcription in prokaryotes and eukaryotes: types of RNA polymerases. DNA elements in transcription: promoters, enhancers, silencers, transcription factors and inhibitors of transcription. Structure of mRNA in prokaryotes and eukaryotes. Post-transcriptional processing in eukaryotes: splicing, capping and polyadenylation. Codon, characteristics of genetic code and wobble hypothesis. Reverse transcription, transposons and retrotransposons. Translation: Mechanism of translation in prokaryotes and eukaryotes: amino acid activation, initiation, elongation, and termination. Posttranslational modification in eukaryotes. Inhibition of protein synthesis by antibiotics.	18
IV	Regulation of Gene Expression: Regulation of Gene Expression in Prokaryotes: Concept of operons, Positive and negative control mechanisms, Lac operon: Inducible system, role of repressor and activator proteins, Trp operon: Repressible system, attenuation mechanism. Regulation of Gene Expression in Eukaryotes-Transcriptional regulation:Transcription factors (activators, repressors), Regulatory sequences: Enhancers, silencers, response elements, Chromatin remodeling (histone modifications, DNA methylation), Hormonal control of gene expression. Post-Transcriptional Gene Regulation: Gene silencing via chromatin modifications, Regulatory RNA:RNA interference (RNAi), MicroRNAs (miRNAs), Riboswitches.	18
	Total	60

Textbooks:

1. Jocelyn E. Krebs; Elliott S. Goldstein; Stephen T. Kilpatrick, Lewin's Genes XII, Jones and Bartlett Publishers, Inc., 12th edition, ISBN: 978-1284104493.
2. Watson, J.D., Molecular Biology of the Gene, Pearsons, 8th edition, ISBN: 978-8176711340.

Reference books:

1. Lodish. H, Berk. A, Lawrence, A, Matsudaira. A, Baltimore. D and Darnell. J. MolecularCell Biology (Fourth Edition). Media Connected – W.H.Freeman and Company. 2009
2. Cell and Molecular Biology by De Robertis E.D.P and De Robertis E.M.F, 1987. LeaandFebiger International Edition, USA. 8thed 2010.
3. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, USA, 7th edition, 2017.
4. Victor Rodwell, David Bender, P. Anthony Weil, Peter Kennelly. Harpers Illustrated Biochemistry 31th Edition, 2018

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom

MAJOR PAPER: Microbiology

Course Level: 300

Subject Code: BCH152M502

L-T-P-C: 3-1-0-4

Scheme of evaluation: Theory (T)

Course Objective

The course will provide the fundamental concepts of Microbiology including structure and classification of bacteria, microbial diversity and pathogenesis

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Remembering key historical developments in microbiology and recall contributions of prominent microbiologists such as Anton von Leeuwenhoek, Louis Pasteur, and Robert Koch.	BT 2
CO 2	Understand and differentiate between prokaryotic and eukaryotic microorganisms and explain the structural and functional differences in microbial cell organization, including Gram-positive and Gram-negative cell walls.	BT1 & BT2
CO 3	Applying their knowledge demonstrate the ability to classify microorganisms based on their nutritional types and growth requirements, and apply techniques for microbial culture, isolation, and growth measurement.	BT3
CO 4	Analyze the mechanisms of microbial pathogenicity and antimicrobial chemotherapy, evaluating the effectiveness of various sterilization and disinfection methods in controlling microbial growth.	BT 5

Course Contents

Modules	Course content	Periods
I	History of Microbiology: History of development of microbiology as a discipline, Spontaneous generation versus biogenesis, contributions of Anton von Leeuwenhoek, Joseph Lister, Paul Ehrlich, Richard Petri, Charles Chamberland, Edward Jenner, Louis Pasteur, Robert Koch, Martinus W. Beijerinck, Sergei Winogradsky, Alexander Fleming, Elie Metchnikoff and Emil vonBehring.	15
II	Diversity of Microbial world and Microbial Cell organization: Difference between prokaryotic and eukaryotic microorganisms. General characteristics of different groups: Acellular microorganisms (Viruses, Viroids, Prions) and Cellular	15

	microorganisms(Bacteria,Archaea,Algae,FungiandProtozoa)withemphasis on distribution, occurrence and morphology. Cell-wall: Composition and detailed structure of Gram positive and Gram's negative cell walls, mechanism of Gram's staining. Cell Membrane: Structure, function and chemical composition of bacterial and archaeal cell membranes.	
III	Microbial Nutrition and Growth: Nutritional types of microorganisms, growth factors, culture media- synthetic and complex, types of media; isolation of pure cultures, growth curves, mean growth rate constant, generation time; influence of environmental factors on growth of microbes: effect of pH, temperature, solute, oxygen concentration, pressure and radiation. Sterilization, disinfection and antiseptics. Use of physical methods (heat, low temperature, filtration, radiation) and chemical agents (phenolics, halogens, heavy metals, sterilizing gases) in microbial control.	15
IV	Pathogenicity of Microorganisms and Antimicrobial Chemotherapy: Introduction to pathogenic microbes; Bacteria, Viruses, Algae, protozoa and fungi. General Characteristics of antimicrobial drugs, determining the level of microbial activity: dilution susceptibility test and disc diffusion test. Range of activity and mechanism of action of penicillin, vancomycin and tetracycline.	15
	Total	60

Textbooks:

1. Lippincott's Illustrated Review of Microbiology by William A Strohl, 5th edition, 2017
2. Willey, J., Sherwood, L., Woolverton, C. (2017). Prescott's Microbiology (10th ed.). McGraw Hill international. ISBN 13:9781259657573

Reference books:

1. Chan, M.J., Krieg E. C. S., Pelczar, N. R. (2004) Microbiology (5th ed.). McGraw Hill International. ISBN 13:9780094623206.
2. Prescott, Harley, Wiley, J.M., Sherwood, L.M., Woolverton, C.J. (2018). Klein's Microbiology. (7th ed.). Mc Graw Hill International Edition (New York) ISBN: 978- 007-126727
3. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, USA, 7th edition, 2017

4. Kathleen parkTalaro (2017). Foundations in Microbiology. 10th Edition, McGraw Hill. Science

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MAJOR PAPER: Molecular Biology and Microbiology Practical

Course Level: 300

Subject Code: BCH152M513

L-T-P-C: 0-0-8-4

Scheme of evaluation: Practical (P)

Course Objective

The objective of the course is to introduce the students to online nucleotide sequence databases and to the basic knowledge/techniques of DNA-isolation, estimation, amplification, and electrophoretic separation.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level

CO 1	Remembering the fundamental principles of DNA isolation, restriction digestion, and electrophoresis techniques.	BT 2
CO 2	Understanding the principles behind DNA quantification, restriction digestion, PCR amplification, and database retrieval of nucleotide sequences.	BT1 & BT2
CO 3	Apply the knowledge and perform genomic and plasmid DNA isolation, quantify DNA using spectrophotometry, and conduct PCR amplification with appropriate primer design.	BT3
CO 4	Analyze, compare and interpret DNA electrophoresis results, assess DNA purity, and analyze retrieved nucleotide sequences for primer design and PCR optimization.	BT 5

Course Contents

Modules	Course content	Periods
I	1. Isolation of genomic DNA from plants/bacteria/yeast 2. Isolation of plasmid DNA from bacteria 3. Quantification and purity determination of isolated genomic DNA by UV-Spectrophotometry.	24
II	1. Basic introduction to nucleotide database 2. Nucleotide sequence retrieval from database 3. Designing of Primer for PCR	24
III	1. Amplification of DNA using PCR 2. Agarose gel electrophoresis of PCR product	24
IV	1. Preparation of Microbial growth curve 2. Gram staining of different bacterial cultures 3. Anti-Microbial sensitivity test	24
	Total	96

Textbooks:

1. Jocelyn E. Krebs; Elliott S. Goldstein; Stephen T. Kilpatrick, Lewin's Genes XII, Jones and Bartlett Publishers, Inc., 12th edition, ISBN: 978-1284104493.
2. Watson, J.D., Molecular Biology of the Gene, Pearsons, 8th edition, ISBN: 978-8176711340.

Reference books:

1. Lodish. H, Berk. A, Lawrence, A, Matsudaira. A, Baltimore. D and Dernell. J. Molecular Cell Biology (Fourth Edition). Media Connected – W. H. Freeman and Company. 2009
2. Cell and Molecular Biology by De Robertis E.D.P and De Robertis E.M.F, 1987. Leaand Febiger International Edition, USA. 8th ed 2010.
3. Biochemistry Ed. Donald Voet& Judith G. Voet, 4thEdn. John Wiley & Sons, Inc.(2012).
4. Introductory Practical Biochemistry – 4th edition, 2017, S.K.Sawhney and Randhir Singh.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

MINOR PAPER: Environmental Biochemistry**Course Level: 200****Subject Code: BCH152N501****L-T-P-C: 3-1-0-4****Scheme of evaluation: Theory (T)****Course Objective**

The course will provide the graduates with the skills needed for environmental stewardship,

sustainable resource management, and informed decision-making for ecological preservation.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Remembering the key components of the environment, including the atmosphere, lithosphere, hydrosphere, and biosphere, and describe their structure and composition.	BT 1
CO 2	Understanding the processes of soil formation, classification of minerals and rocks, and the global water balance, highlighting their role in ecosystem sustainability.	BT2
CO 3	Analyze the impact of human activities on natural resources, biodiversity, and environmental pollution, proposing practical measures for conservation and sustainable development.	BT3
CO 4	Evaluate the interconnections between ecosystems, biogeochemical cycles, and environmental pollution, assessing the effectiveness of bioremediation and sustainable management practices in mitigating environmental issues.	BT 4

Course Contents

Modules	Course content	Periods
I	Environmental components: Atmosphere, structure and chemical composition of atmosphere, Internal structure of the Earth. Global Water Balance. Hydrological cycle. Classification of trace elements, mobility of trace elements, biogeochemical cycles. Renewable and non-renewable resources. Definition, scope and importance, need for public awareness. Natural resources, types, uses and overexploitation. Role of an individual in conservation of natural resources.	15
II	Fundamentals of ecology: Ecosystems, Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem, Ecological succession, Food chains, food webs and ecological pyramids. Introduction, to types of ecosystems and their characteristic features. Biodiversity and its conservation	15

III	Environmental Issues: Environmental Pollution Definition, Cause, effects and control measures of: Air pollution, Water pollution, Soil pollution, Solid waste Management: Causes, effects and control measures of urban and industrial wastes, Role of an individual in prevention of pollution, Pollution case studies.	15
IV	Sustainable development: Sustainable development and sustainable goals. Bioremediation: Introduction and types of bioremediation, bioremediation of surface soil and sludge, bioremediation of subsurface material, In situ and Ex-situ technologies, Phytoremediation.	15
	Total	60

Textbooks:

1. Environmental Biochemistry – NeelimaRajvaidya, Dilip Kumar Markandey(2005).
2. Environmental and Ecological Biochemistry P.W. Hochachka, T.P. Mommsen

Reference books:

1. Environmental Biochemistry Hardcover – 2005 by D. K. Markandey, N.Rajvaida.
2. Fundamentals of Ecology 5th Edition by Eugene Odum, Gary W. Barrett.
3. Environmental Chemistry Paperback –by V. K. Ahluwalia, Lalita S. Kumar, ANE Books.
4. **Baird, C., & Cann, M.** (2012). *Environmental chemistry* (5th ed.). W. H. Freeman.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

Syllabus 6th semester

MAJOR PAPER: ADVANCED CELL BIOLOGY

Subject Code: BCH152M601

L-T-P-C: 2-1-0-3

Scheme of evaluation: Theory (T)

Course Objective

To provide knowledge about different concepts of cell biology

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Recall and remember the basic information about cellular organelles between cells, tissues and organs.	BT 1
CO 2	Define and describe different aspects of chromosomal organization	BT1 & BT2
CO 3	Discover the modes of intracellular transport	BT3
CO 4	Analyze different mechanisms of cellular functions	BT 4

Modules	Course content	Periods
I	Basic concepts of cell biology: Evolution of cells, cell architecture, structural organization of prokaryotic and eukaryotic cells. Structure and functions of different cell organelles: nucleus, mitochondria, chloroplast, rough and smooth endoplasmic reticulum, Golgi apparatus, lysosomes, and peroxisomes. Cytoskeleton: actin,	15

	microfilaments, intermediate filaments, and microtubules. Subcellular fractionation and Svedberg unit.	
II	Chromosomal organization, cell cycle and cell division: Definition of a gene, chromosomal organization of gene; Types of chromosomes: autosomes, sex chromosomes; Karyotyping; Chromatin, Histones and Nucleosomes; Heterochromatin and euchromatin; Structural chromosomal abnormalities: Chromosome ploidy, Polyploidy and aneuploidy. Cell cycle and its regulation; cell cycle checkpoints Events of mitosis and cytokinesis; Events of meiosis, Regulation of cell division.	15
III	Vesicular Transport Intracellular trafficking: Understanding the transport of proteins, lipids, and macromolecules between organelles and the plasma membrane. Exploration of endocytosis and exocytosis, including their stepwise processes such as vesicle trafficking, tethering, docking, fusion, and release. Mechanisms of Endocytosis: Phagocytosis (cell eating), Pinocytosis (cell drinking) and Receptor-mediated endocytosis. Role of Golgi apparatus, Endoplasmic Reticulum (ER) and Lysosomes in protein sorting, packaging, and degradation.	15
IV	Signal Transduction Pathways: Types of cell signalling: autocrine, paracrine, and endocrine signalling; Cell surface receptors and intracellular receptors; Mechanisms of ligand binding, receptor activation, and conformational changes. Introduction to second messengers like cAMP and calcium; Role of second messengers in signal transduction; Activation of G Protein-Coupled Receptors (GPCRs) and receptor tyrosine kinase (RTKs); signalling pathways Ras-MAPK, PI3K/Akt/mTOR, JAK-STAT pathway. Mechanism of cell death pathways: Apoptosis, Necrosis etc.	15
Total	60	60

Text Books:

1. Molecular Biology of the Cell, 6th Edition. Bruce Alberts, Dennis Bray, Julian Lewis,
2. Martin Raff, Keith Roberts, and James D Watson. Publisher New York: Garland Science.
3. The Cell: A Molecular Approach, 7th Edition, by Geoffrey M. Cooper and Robert E. Hausman, published by ASM Press.

Reference:

1. Lipid Biochemistry; 5th Edn. Michael I. Gurr, John L. Harwood and Keith N. Frayn, Blackwell Science (2002).
2. Principles of Human Physiology; 6th Edn. Cindy L. Stanfield Pearson, (2011).

3. Biochemistry Ed. Donald Voet& Judith G. Voet, 4thEdn. John Wiley & Sons, Inc. (2012).
4. Mammalian Biochemistry; White, Handler and Smith, McGraw-Hill,(1986)

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MAJOR PAPER: MEMBRANE BIOLOGY AND BIOENERGETICS Subject Code: BCH152M602 L-T-P-C: 2-1-0-3 Scheme of evaluation: Theory (T)
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Course Objective

To provide knowledge about different concepts of cell biology

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Students will study about the structure of bio-membranes, the fluid mosaic model, and the electron transport chain	BT 1
CO 2	They will summarize different aspects of biomembranes	BT 2
CO 3	They will analyze the aspects of membrane dynamics	BT3
CO 4	They will compare different transport processes across the membrane	BT 4

Detailed Syllabus:-

Modules	Course content	Periods
I	Principles of Bioenergetics: Energy transformation, Laws of thermodynamics and biological relevance. Entropy, Enthalpy, Gibbs energy, free energy changes, and redox potentials. High energy compounds: examples and structures. Reducing equivalents and electron transport chains. Chemiosmotic Theory: Proton gradient, proton motive force and ATP synthesis. ETC inhibitors and uncouplers.	15
II	Introduction to Biomembranes: Composition of biomembranes - prokaryotic, eukaryotic, neuronal, and subcellular membranes. Study and classification of membrane proteins and lipids. Experimental validation of Fluid mosaic model with experimental proof. Factors determining the membrane fluidity. Different properties of the Phospholipid Bilayer. Role of Cholesterol in Membrane Structure.	15
III	Membrane Dynamics and Structures: Polymorphic structures of amphiphilic molecules in aqueous solutions - micelles and bilayers. C.M.C., Critical Packing parameter and interactions stabilizing the membrane structure. Membrane asymmetry and lipid rafts. Techniques for studying membranes: freeze-fracture electron microscopy, fluorescence recovery after photobleaching (FRAP).	15
IV	Membrane transport: Thermodynamics of transport. Simple diffusion and facilitated diffusion. Passive transport - glucose transporter, anion transporter and porins. Primary active transporters - P type ATPases, V type ATPases, F type ATPases. Secondary active transporters - lactose permease, Na ⁺ -glucose symporter. A.B.C. family of transporters - M.D.R. and CFTR. Group translocation. Ion channels - voltage-gated ion channels (Na ⁺ /K ⁺ voltage-gated channel), ligand-gated ion channels (acetyl choline receptor), aquaporins, and bacteriorhodopsin. Ionophores - valinomycin, gramicidin.	15
Total	60	60

Text Books:

1. *Lehninger Principles of Biochemistry*, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, U.S.A. 7th edition, 2017
2. *Harper's Illustrated Biochemistry*; 31st Edn. Robert K. Murray, Daryl K. Granner, Victor

3. W. Rodwell, the McGraw-Hill (2018).
4. Biochemistry by Donald Voet and Judith Voet, John Wiley and Sons NY. 4thedtn, 2012.

Reference:

1. Berg, J. M., Tymoczko, J. L. and Stryer. *Biochemistry*, W.H Freeman and Co., 9th Edition, 2019.
2. Buchanan, B., Gruissem, W. and Jones, R., *Biochemistry and Molecular Biology of Plants*, 2nd Edn, American Society of Plant Biologists, U.S.A.
3. Fundamentals of Biochemistry by J.L. Jain, Nitin Jain and Sunjay Jain, S. Chand Group.
4. Biochemistry Ed. Donald Voet & Judith G. Voet, 4th Edn. John Wiley & Sons, Inc. (2012).

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MAJOR PAPER : Immunology Subject code - BCH152M603 Credit: 3-1-0-4 Scheme of Evaluation: Theory (T)
Course Objectives:
The course will provide the basic framework in immunology that will cover the major topics including innate and adaptive immunity, antibodies and antigens, and the molecular events leading to autoimmunity.
Course outcomes-
On Successful Completion of the course the students will be able to:

Sl No	Course Outcome	Blooms Taxonomy Level
CO 1	Demonstrate the basic knowledge of immunological processes at a cellular and molecular level.	BT3
CO 2	Compare and contrast the key mechanisms and cellular players of innate and adaptive immunity and how they are related	BT4
CO 3	Identify the genetic basis for immunological diversity and the generation of immune responses, and molecular basis of allergic and inflammatory responses	BT2
CO4	Relate the basis of immunological tolerance, autoimmunity and transplantation and immune dysregulation in various disorders	BT4

Course contents:

Modules	Topics / Course content	Periods
I	Introduction to immune system Introduction to the cells and organs of immunity; innate and adaptive immune system, cell mediated and humoral immunity, B and T lymphocytes, primary and secondary immune response, passive and active immunity, clonal selection theory, Antigens and immunogens. Structure and functions of immunological molecules: Ig, TCR, MHC molecules, CD3, CD4 and CD8	15
II	Immunological mechanisms- Genetic basis of antibody diversity; Immunological co-operations; Antigen processing and presentation for T cells; Acquired immunological tolerance; Complement activation; Apoptosis; Interleukins and interferons.	15

III	Immune system in health and disease- Hypersensitivity; Tumor immunology; Immunodeficiency and autoimmunity; Transplant rejection reactions; Vaccines.	15
IV	Measurement of antigen-antibody interactions- Direct binding assays, agglutination and precipitation, immunoelectrophoresis, complement fixation reaction, radioimmunoassay, and ELISA, monoclonal antibodies.	15
Total		60

Text Books:

1. Kuby Immunology; Owen, Punt, Stranford, 8thEdn. W. H. Freeman (2018).
2. Roitt's Essential Immunology; Ivan, M. Rohitt&Petrer J Delves 13th (2011) Blackwell Science.

Reference Books:

1. Antibodies– A Laboratory Manual; E. D. Harlow, David Lane, 2nd Edn. CSHL Press (2014).
2. Basic and Clinical Immunology; Stites et al., [6thedn] (2011) Lange.
3. Veterinary Immunology: Ian R. Tizard, 10thEdn. I.R. Thomson press.
4. Biochemistry Ed. Donald Voet& Judith G. Voet, 4thEdn. John Wiley & Sons, Inc.(2012).

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MAJOR PAPER : Immunology and Cell Biology Practical Subject code: BCH152C613 Credit:0-0-8-4		
Course objectives The course will provide the basic laboratory exposure to the students in the field of immunology		
Course outcomes-		
On Successful Completion of the course the students will be able to:		
Sl No	Course Outcome	Blooms Taxonomy Level
CO 1	Understand the basic immunology lab approaches	BT1
CO 2	Demonstrate the ability to use discipline specific research techniques	BT3
CO 3	Apply their skills to conduct and execute experiments	BT3
CO4	Explain the experimental results on the basis of critic analysis	BT2

Course contents:

Modules	Topics / Course content	No. of hrs
I	1. Separate serum from blood 2. Separate plasma from blood sample	24
II	1. To determine the blood groups of humans by A, B, O blood group system 2. Single radial immunodiffusion assay	24
III	1. Mitosis in onion root tip/ insect gonads. 2. Meiosis in onion root tip/insect gonads	24
IV	1. Isolation of lymphocytes from blood 2. Preparation of metaphase slides from lymphocytes	24
Total		96

Text Books:

1. Antibodies– A Laboratory Manual; E. D. Harlow, David Lane, 2nd Edn. CSHL Press(2014).
2. Basic and Clinical Immunology; Stites et al., [6thedn] (2011)Lange.

Reference Books:

1. Kuby Immunology; Owen, Punt, Stranford, 8thEdn. W. H. Freeman (2018).
2. Veterinary Immunology: Ian R. Tizard, 10thEdn. I.R. Thomson press.
3. The Immune System. By Peter Parham 4thEdn. Publisher Garland publishing
4. Ashim, K. C. (2014). *Experimental immunology: Essential methods*. Wiley-Blackwell.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

Minor paper: Basics of Genetic Engineering Subject code: BCH152N601 Credit: 3-1-0-4 Student's scheme of evaluation: Theory paper		
Course Objectives		
The objective of the course is to teach the basics of theoretical aspects of recombinant DNA technology and genetic engineering		
Course outcomes		
On Successful Completion of the course the students will be able to:		
Sl. No.	Course Outcome	Bloom's Taxonomy level
CO1	Define and comprehend the fundamental concepts of genetic engineering and biotechnology	BT1
CO2	Understand the process of recombinant DNA technology and tools that are vital for the same	BT2
CO3	Apply the knowledge of genetic engineering and biotechnology to comprehend how these are instrumental in the fields of agriculture, medicine, and pharmaceuticals	BT3
CO4	Explain and analyze the experimental results and the ethical concerns associated with the use of genetic engineering in manipulating the genome of organisms for human benefit	BT4

Course Contents:

Module	Topics/Course Contents	Periods
I	Introduction to Genetic Engineering and Biotechnology Overview of recombinant DNA technology, Biotechnology, and Genetic Engineering, Significance of Genetic Engineering in Agriculture and Medicine, Bioethics.	15

II	Molecular Genetics Tools Isolation of Genomic DNA from Bacteria, Plant, Fungal and Animal cells, Extraction of Plasmid DNA, Restriction Enzymes, Restriction and modification systems, DNA ligase, Extraction and purification of plasmid and bacteriophage DNA, Separation of DNA by gel electrophoresis.	15
III	Molecular Cloning Cell-based cloning and Cell-free cloning, Gene transfer technologies (Direct vs indirect), Vectors for cloning, Plasmid as vectors (pBR322 and pUC8), <i>Agrobacterium</i> as a natural genetic engineer.	15
IV	Polymerase chain reaction and DNA sequencing Fundamentals of polymerase chain reaction, designing primers for PCR. Studying PCR products. Cloning PCR products. Real-time PCR, DNA sequencing	15
	Total Periods	60

Textbook

1. Brown, T. A. Gene Cloning and DNA analysis: an introduction. Blackwell Science, 6th edition 2010.
2. Nicholl, D.S.T. (2008) An introduction to Genetic Engineering (3rd edition) Cambridge University Press.

Reference Book

1. Watson, J.D. (2006) Recombinant DNA (3rd edition) Cold Spring Harbor Laboratory Press.
2. Primrose, S.B. and Twyman, R. (2006) Principles of Gene Manipulation and Genomics (7th edition) Blackwell Publishing
3. Glick, B. R., Pasternak, J. J., & Patten, C. L. (2017). *Molecular biotechnology: Principles and applications of recombinant DNA* (5th ed.). ASM Press.
4. Kreuzer, H., & Massey, A. (2012). *Recombinant DNA and biotechnology: A guide for students* (3rd ed.). ASM Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

Syllabus 7th semester

MAJOR PAPER : Genetic Engineering and Biotechnology Subject code: BCH152M701 Credit: 3-1-0-4 Student's scheme of evaluation: Theory paper		
Course Objectives		
The objective of the course is to teach the basics of theoretical aspects of recombinant DNA technology and genetic engineering		
Course Outcome		
On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Bloom's Taxonomy level
CO1	Define and comprehend the fundamental concepts of genetic engineering and biotechnology	BT1
CO2	Understand the process of recombinant DNA technology and tools that are vital for the same	BT2

CO3	Apply the knowledge of genetic engineering and biotechnology to comprehend how these are instrumental in the fields of agriculture, medicine, and pharmaceuticals	BT3
CO4	Explain and analyze the experimental results and the ethical concerns associated with the use of genetic engineering in manipulating the genome of organisms for human benefit	BT4

Course Contents:

Module	Topics/Course Contents	Periods
I	Introduction to recombinant DNA technology- Overview of recombinant DNA technology. Restriction and modification systems, restriction endonucleases and other enzymes used in manipulating DNA molecules, and separation of DNA by gel electrophoresis. Extraction and purification of plasmid and bacteriophage DNA.	15
II	Cloning vectors for prokaryotes and eukaryotes- Plasmids and bacteriophages as vectors for gene cloning. Cloning vectors based on <i>E. coli</i> plasmids, pBR322, pUC8, pGEM3Z. Cloning vectors based on M13 and λ bacteriophage. Vectors for yeast, higher plants and animals. The problem of selection, direct selection, marker rescue. Gene libraries, identification of a clone from gene library, colony and plaque hybridization probing, methods based on detection of the translation product of the cloned gene.	15
III	Polymerase chain reaction and DNA sequencing Fundamentals of polymerase chain reaction, designing primers for PCR. Studying PCR products. Cloning PCR products. Real-time PCR, DNA sequencing by Sanger's method, and modifications based on Sanger's method. Automated DNA sequencing. Pyrosequencing.	15

IV	Applications of Genetic Engineering in Biotechnology Site-directed mutagenesis and protein engineering. Applications in medicine: production of recombinant pharmaceuticals such as insulin, human growth hormone, Factor VIII, Recombinant vaccines, and Gene therapy. Applications in agriculture: plant genetic engineering, herbicide-resistant crops, problems with genetically modified plants, safety concerns.	15
	Total Periods	60

Textbook

1. Brown, T. A. Gene Cloning and DNA analysis: an introduction. Blackwell Science, 6th edition 2010.
2. Nicholl, D.S.T. (2008) An introduction to Genetic Engineering (3rd edition) Cambridge University Press.

Reference Book

1. Watson, J.D. (2006) Recombinant DNA (3rd edition) Cold Spring Harbor Laboratory Press.
2. Primrose, S.B. and Twyman, R. (2006) Principles of Gene Manipulation and Genomics (7th edition) Blackwell Publishing
3. Glick, B. R., Pasternak, J. J., & Patten, C. L. (2017). *Molecular biotechnology: Principles and applications of recombinant DNA* (5th ed.). ASM Press.
4. Kreuzer, H., & Massey, A. (2012). *Recombinant DNA and biotechnology: A guide for students* (3rd ed.). ASM Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MAJOR PAPER : Research Methodology, Bioethics, and Scientific Writing Subject code: BCH152M702 Credit: 3-1-0-4 Student's scheme of evaluation: Theory paper		
Course Objectives		
The objective of this course is to equip students with essential skills in research methodology, bioethics, and scientific writing to design, conduct, and communicate scientific research effectively		
On Successful Completion of the course the students will be able to:		
Sl. No.	Course Outcome	Bloom's Taxonomy level
CO1	Understand basic research methods and design research projects.	BT1
CO2	Understand and apply ethical principles in scientific research.	BT1/BT2
CO3	Apply the knowledge of statistical analysis in critically evaluating the importance of scientific experimentation and data visualization .	BT3
CO4	Critically read, review scientific research papers, and write clear and organized scientific papers using proper formats	BT4

Course Contents:

Module	Topics/Course Contents	Periods
I	Research Methodology: Introduction to Research: What is research? Types of research (Qualitative vs. Quantitative). Research Design: Basic designs (e.g., experimental, survey). Data Collection: Surveys, interviews, observations. Research Ethics: Ethical guidelines in research (informed consent, confidentiality).	15
II	Bioethics in Research Introduction to Bioethics: Key principles (autonomy, justice). Ethical Issues: Human and animal research, informed consent. Case Studies: Real-life examples of bioethical issues in research.	15

III	Basics of Biostatistics Statistics in research: Concepts of sample and population, Measure of central tendency, dispersion, asymmetry (skewness, kurtosis), Normal distribution (p-value), Statistical tests and hypothesis (Standard error, t-test, chi-square test), and regression analysis	15
IV	Scientific Writing, Critical Reading. and Review of Scientific Literature Writing a Research Paper: Introduction, methods, results, and discussion (IMRAD format).Citing Sources: Importance of citations, avoiding plagiarism.Figures and Tables: How to present data visually. Critical Reading: How to analyze scientific papers.Evaluating Research: Assessing quality, methods, and results.Writing a Literature Review: Summarizing key findings from articles. Scientific report writing.	15
	Total Periods	60

Textbook

1. Research in Education (1992) 6th ed. Best, J.W. and Kahn, J.V., Prentice Hall of India Pvt. Ltd. ISBN-978-81-203-3563-9.
2. AttheBench:ALaboratoryNavigator(2005)Barker,K.,ColdSpringHarborLaboratoryPress (New York), ISBN: 978-087969708-2.42

Reference books:

1. Research Methodology - Methods and Techniques (2004) 2nd ed., Kothari C.R., New Age International Publishers. ISBN –81-224-1522-9
2. Research Methodology: A Step-by-Step Guide for Beginners (2005) 2nd ed. Kumar R., Pearson Education. ISBN: 978-1-4129-6467-8.
3. Biostatistics: A Foundation for Analysis in the Health Sciences (2013) 10th ed., Daniel W.W., John Wiley and Sons Inc. ISBN-13: 978-1118302798 ISBN-10:1118302796
4. Biostatistics at the Bench: A Step-by-Step Handbook for Biologists (2010) Bremer, M. and Doerge, R.W., Cold Spring Harbor Laboratory Press (New York), ISBN: 978-0-879698-57-7.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

MAJOR PAPER : Plant Biochemistry		
Subject code: BCH152M703		
Credit: 3-1-0-4		
Student's scheme of evaluation: Theory paper		
Course Objectives		
The objective of this course is to provide a deep understanding of the cellular structure and insights into metabolic processes in plants.		
Course Outcome		
On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Bloom's Taxonomy level
CO1	Able to gain knowledge of basic plant cell structure, photosynthesis, carbon cycle, stress responses, and plant defenses.	BT1
CO2	Understand the basic makeup of cell structure, the scheme of photosynthesis, the reactions of carbon cycles, the hormones and metabolites in stress, and pathogen response.	BT1/BT2
CO3	Analyze the functioning of the plant cell metabolism under various conditions.	BT3
CO4	Evaluate the metabolites and biochemical pathways that are critical for plant survival.	BT4

Course Contents:

Module	Topics/Course Contents	Periods
I	Introduction to plant cell structure and photosynthesis: Introduction to Plant cells, Plasma membrane, Vacuole and Tonoplast membrane, Cell wall, Plastids and Peroxisomes. Molecules involved in photosynthesis: chlorophyll and carotenoids. Antenna molecules. Structure of P.S.I. and PSII complexes. Photosynthetic reactions and the Z scheme. Role of the P.S.I. and PSII in photosynthetic reactions. Regulation of P.S.I. and PSII activity. Functioning of oxygen-evolving complex.	15
II	Carbohydrate biosynthesis in plants: CO ₂ fixation- stages and regulation. Energetic requirements and light-driven enzymes of Calvin cycle. Photorespiration, C ₄ and C.A.M. pathways. Salvage pathway. Rubisco activity in C ₄ and C.A.M. plants. Biosynthesis of starch and sucrose and its regulation. Synthesis of cellulose.	15
III	Regulation of plant growth and stress physiology: Introduction to plant hormones. Biosynthesis of plant hormones- site and pathways of synthesis, transport, and mode of action. Plant responses to abiotic and biotic stresses, Water deficit and drought resistance, Flooding, Temperature stress, Salt stress.	15
IV	Secondary metabolites and plant defense: Cutin, waxes, lectins and suberin- structure, and role in plant defense. Secondary metabolites- biosynthetic pathways and types. Terpenes- biosynthesis, roles in plant growth and defense. Phenolic compounds- types, biosynthesis, activation, functions in plant growth and defense. Role of amino acids, proteins and hormones in plant defense. Response of plants to infections	15
	Total Periods	60

Textbook

1. Plant Physiology by Taiz, L. and Zeiger, E. (2015). (6th Ed.).
2. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, U.S.A. 7th edition, 2017

References

1. Buchann (2015). Biochemistry and Molecular Biology of Plant. (2nd Ed.).
2. Caroline Bowsher, Martin Steer, Alyson Tobin (2015). Plant Biochemistry. Garland Science.
3. Dey, P. M. and J.B. Harborne, J.B., (Editors) (2013). Plant Biochemistry.
4. Heldt, H.-W., & Piechulla, B. (2021). *Plant biochemistry* (5th ed.). Academic Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

Major paper: Plant Biology and Genetic Engineering Practical Subject code: BCH152M713 Credit: 0-0-8-4 Scheme of evaluation: Practical
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Course Objective

The objective of this paper is to offer insights about the practicals based on analysis of biomolecules as well on the cell biology.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Describe the working principles of various methods to prepare plant	BT1

	extracts	
CO 2	Explain the principles behind utilizing different chemicals in plant extract preparation.	BT2
CO 3	Demonstrate appropriate methods for detecting polyphenols and flavonoids in plant extracts.	BT3
CO 4	Compare the efficiency of different solvents in plant extract preparation.	BT4

Course Contents

Modules	Course content	Periods
I	Preparation of methanolic plant extract Preparation of ethanolic plant extract Preparation of aqueous plant extract	20
II	Estimation of soluble sugars in plant extract Estimation of pyruvate in plant extract Lectin assay by haemagglutination	26
III	Assay of free radical scavenging activity of plant extracts Detection of polyphenols in plant extract Estimation of flavonoids in plant extract	24
IV	Isolation of plasmid DNA from bacteria To perform restriction digestion of DNA Agarose gel electrophoresis of DNA	20
Total		90

Textbooks:

1. An Introduction to Practical Biochemistry – 3rd edition, 2017, David Plummer.
2. A practical manual on Fundamentals of Plant Biochemistry and Biotechnology. 1st Edition, 2023, KK Panigrahi, 2023, College of Agriculture (OUAT).

Reference books:

1. Practical in plant physiology and Biochemistry. Manju Bala, 2024. Scientific Publishers
2. Introductory Practical Biochemistry – 4th edition, 2017, S.K. Sawhney and Randhir Singh..

3. **Jain, J. L., Jain, S., & Jain, N.** (2016). *Fundamentals of plant physiology and biochemistry: Practical manual*. S. Chand Publishing.
4. **Verma, S. K., & Verma, M.** (2007). *A textbook of plant physiology, biochemistry, and biotechnology*. S. Chand Publishing.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
0 hours	90 hours	0 hours

MINOR PAPER: Cancer Biology Subject code: BCH152N701 Credit: 3-1-0-0 Scheme of evaluation: Theory		
Course Objectives		
The objective of the course is to provide a brief idea about cancer, its causes, and available therapeutic options		
Course Outcome		
On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Bloom's Taxonomy level
CO1	Understand the fundamental differences between cancerous and normal cells, including definitions, types, and the roles of oncogenes and tumor suppressor genes	BT1
CO2	Understand the mechanisms of cell cycle regulation and cell death pathways, such as apoptosis, necrosis, and autophagy, in the context of cancer development	BT2

CO3	Apply knowledge of carcinogenic agents and DNA repair mechanisms to assess factors contributing to carcinogenesis	BT3
CO4	Evaluate various cancer treatment strategies, including traditional therapies like surgery, chemotherapy, radiation, as well as emerging treatments, considering their challenges and effectiveness.	BT4

Course Contents:

Module	Topics/Course Contents	Periods
I	Introduction to Cancer Definition of cancer, cancerous cell vs normal cells, cancer types (benign and malignancies), oncogenes and tumor suppressor genes, and carcinogenesis.	15
II	Cell Cycle and Cell Death Pathways in Cancer Cell cycle and checkpoint, premature anaphase separation, genomic instability, Tumour suppressor and cell cycle, Cell death pathways and cancer -Apoptosis, Necrosis, and Autophagy	15
III	Carcinogenic agents and Carcinogenesis Mutagenesis vs Carcinogenesis; Carcinogens and their types; Life-style factors responsible for Cancer; Chromosomal instability and its role in cancer; insults to DNA and basics of repair mechanisms.	15
IV	Cancer Treatment Strategies Risk factors: Genetic, environmental, and lifestyle, Cancer prevention strategies: Diet, exercise, screenings, Traditional cancer therapies: Surgery, chemotherapy, radiation therapy, Targeted therapy and immunotherapy, Drug resistance and challenges in cancer therapy, Emerging treatments:	15
	Total Periods	60

TEXT BOOKS:

1. Weinberg, R.A. "The Biology of Cancer" Garland Science, 2013, ISBN- 978-0815342205
2. Devlin T. M. (2011) Textbook of Biochemistry with clinical correlations, Wiley-Liss Publ.

Reference books-

1. Hall, J.E. (2015), Guyton and Hall Textbook of Medical Physiology, Elsevier.
2. Litwack G. (2008) Human Biochemistry and Disease, Academic Press, Elsevier Inc.
3. Molecular biology of cancer: Mechanisms, Targets and therapeutics. Pecorino L, Oxford University Press, 2021, ISBN- 978-0198833024

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Presentations

SYLLABUS 8th SEMESTER

MAJOR PAPER: Bioanalytical techniques

Subject Code: BCH152M801

L-T-P-C: 3-1-0-4

Scheme of evaluation: Theory (T)

Course Objective

This course aims to introduce students to various techniques and their underlying principles utilized in biological research, emphasizing practical skills so that students may apply this information to increase their comprehension of the subject and execute these techniques more effectively.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Learn and understand the principles and applications of spectrophotometry and fluorimetry.	BT1 & BT2
CO 2	Understand the fundamentals of several chromatographic procedures, such as gel filtration and ion exchange.	BT2
CO 3	Apply the knowledge of electrophoretic methods and principles in protein and nucleic acid analysis.	BT3
CO 4	Apply the principles of sedimentation and compare the various types of centrifuges and rotors.	BT3 & BT4

Course Contents

Modules	Course content	Periods
I	Electrophoretic techniques: Principles of electrophoretic separation. Continuous, free, zonal and capillary electrophoresis, different types of electrophoresis including paper, cellulose, pulse field gel electrophoresis, 2D gel electrophoresis. Spectroscopy: Concepts of spectroscopy, Visible and UV spectroscopy, Laws of photometry. Beer-Lambert's law, Principles and applications of colorimetry.	15
II	Chromatography: Principles of partition chromatography, paper, thin layer, ion exchange and affinity chromatography, gel permeation chromatography, HPLC, gas chromatography. LC-MS, GC-MS	15

III	Centrifugation: Principles of centrifugation, concepts of RCF, different types of instruments and rotors, preparative, differential and density gradient centrifugation, analytical ultra centrifugation, determination of molecular weights and other applications, sub cellular fractionation	15
IV	Electron microscopy: Light, electron (scanning and transmission), phase contrast, fluorescence microscopy, freeze-fracture techniques, specific staining of organelles or marker enzymes.	15
	Total	60

Textbooks:

1. Principles and Techniques of Practical Biochemistry, Keith Wilson and John Walker, 7th Edition, 2010.
2. Physical Biochemistry, application to Biochemistry and Molecular Biology, David Freifelder, 10th Edition, 2018.

Reference books:

1. R. F. Boyer, Modern experimental biochemistry, Benjamin Cummings, San Francisco, 5th ed., 2016.
2. I D. Harvey, Modern analytical chemistry, McGraw-Hill, Boston, 2019.
3. Upadhyay, A., Upadhyay, K., & Nath, N. (2016). *Biophysical chemistry: Principles and techniques* (4th ed.). Himalaya Publishing House.
4. **Price, N. C., & Dwek, R. A.** (2010). *Principles and problems in physical chemistry for biochemists* (3rd ed.). Oxford University Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Seminar Flipped classroom

MINOR PAPER: Microbial Ecology

Subject Code: BCH152N801

L-T-P-C: 3-1-0-4

Scheme of evaluation: Theory (T)

Course Objective

The objective of the course is to teach the basics of microbial ecology and their role in the ecology of the environment.

Course Outcome

On Successful Completion of the course, the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Recognize the different types of microbial flora	BT1
CO 2	Summarize the various food webs in the microbial world	BT 2
CO 3	Analyse the role of microbes in maintaining soil health and agriculture yields.	BT 3
CO 4	Analyse and apply the role of microbes in mitigating environmental pollution	BT 3 & BT4

Course Contents

Modules	Course content	Periods
I	Whittaker's Five-kingdom and three-kingdom concept of living organisms (General characteristics of those groups); General features of Eubacteria and Archaeobacteria (major difference with Eubacteria). Structure and function of ecosystems, Terrestrial Environment: Soil profile and soil microflora, Aquatic Environment: Microflora of fresh water and marine habitats, Atmosphere: Aeromicroflora and dispersal of microbes, Animal Environment: Microbes in/on human body (Microbiomics) & animal (ruminants) body.	12
II	Definitions of growth, measurement of microbial growth, Batch culture, Continuous culture, Microbial growth in response to environment -Temperature (psychrophiles, mesophiles, thermophiles, extremophiles, thermodurics, psychrotrophs), pH (acidophiles, alkaliphiles), solute and water activity (halophiles, xerophiles, osmophilic), Oxygen (aerobic, anaerobic, microaerophilic, facultative aerobe, facultative anaerobe), barophilic. Microbial growth in response to nutrition and energy – Autotroph/Phototroph, heterotrophy, Chemolithoautotroph, Chemolithoheterotroph, Chemoheterotroph, Chemolithotroph, photolithoautotroph, Photoorganoheterotroph. Different types of microorganisms in the air, aerosols, sampling techniques, airborne pathogens, techniques of room sterilization.	12
III	Physical and chemical characteristics of various soil types-different microbial groups in soil, method of study,. Rhizosphere, Phyllosphere. Brief account of microbial interactions - (symbiosis, neutralism, commensalism, competition, ammensalism, synergism, parasitism, and predation); Biological nitrogen fixation - symbiotic and asymbiotic; Root - nodule formation in legumes; Biogeochemical Cycles-Carbon, Nitrogen, Phosphorus, and Sulphur Cycles –role of micro organisms in the process, Microbes-mediated key events such as Nitrogen fixation,	12

	ammonification, nitrification, denitrification and nitrate, reduction.	
IV	Principles of Coliform test - detection of faecal and non-faecal coliform); IMViC test; Environmental application of microbes in waste management: Sources and types of solid waste and liquid waste, contribution of the microbial load, Methods of solid waste disposal (composting and sanitary landfill), Microbiological treatment of sewage and industrial waste water: Composition and strength of sewage (BOD and COD), Primary, secondary (oxidation ponds, trickling filter, activated sludge process and septic tank) and tertiary sewage treatment. Major waterborne pathogens, diseases and mode of sterilization.	12
	Total	48

Textbooks:

- 1 Madigan M. T., Martinko J. M., Stahl D. A. and Clark D. P. (2017) Brock Biology of Microorganisms. 14th edition, Pearson
- 2 Prescott L.M., Harley J.P. and Klein D.A. (2007) Principles of microbiology, Wm C. Brown Publ

Reference books:

- 1 Tortora G.J. et al (2015) Microbiology: An Introduction, 12th edition. The Benjamin / Cummings Pub. Co.
- 2 Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M., WH Freeman and Company, New York, USA. 7th edition, 2017
- 3 Subba Rao, NS. Soil Microbiology, 4th Ed., Oxford & IBH Publishing Co. Pvt.
- 4 **Konopka, A.** (2009). *Microbial ecology: Contemporary advances and applications* (1st ed.). Wiley-Blackwell.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment

		Group discussion Flipped classroom
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Subjects in lieu of Dissertation

Subject name: Virology		
Subject code:		
Credits: 3-1-0-4		
Scheme of evaluation: Theory		
Course Objectives:		
The Course aims to enable the students to understand the basics of learning virology and to impart knowledge on the implications of human viral diseases and newer emerging viral infections		
Course Outcome		
On Successful Completion of the course the students will be able to:		
Sl No	Course Outcome	Blooms Taxonomy Level
CO 1	Define the basics of Virology	BT1
CO 2	Recognise the classification and the diversity of viral genomes	BT 1
CO 3	Describe the various replication strategies employed by DNA and RNA viruses	BT 2 & 3
CO4	Analyse anti-viral therapies and the patterns of emergence of new	BT3

	epidemics	
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Course contents:

Modules	Topics / Course content	Periods
I	General perspectives: History and Origin: the development of the concept of viruses as living and nonliving entities. General characteristics of viruses: What are viruses? Difference between bacteria and viruses, Components of viruses, sizes, and shapes of different viruses (describe with at least one example).	15
II	Classification of viruses based on the nucleic acid content: DNA (dsDNA, ssDNA) and RNA (ssRNA, dsRNA) viruses with examples. Human cancer viruses (SV40, HTLV - 1 & 2, Epstein-Barr virus only) Virus-like agents: viroids & prions.	15
III	Viral replication: General characteristics of replication, Replication of T4 phage. Phage growth and the estimation of phage numbers. The lytic and lysogenic life cycle of bacteriophage lambda; mechanism(s) that determines lytic and lysogenic life cycle. Reverse transcription and integration by Retroviruses.	15
IV	Viral pathogenesis and cell transformation by viruses. Viral vaccines, Antiviral therapy. The emergence of new viruses and Current viral epidemics with emphasis on SARS-Cov2 and H1N1 Swine Flu, Viral vectors, and gene therapy.	15
	TOTAL	60

Text Books:

1. Principles of Virology – S J Flints, L W Enquist, R M Krug, V R Racaniello, A M Skalka. ASM Press. 5th edition, 2020
2. Field Virology – David M. Knipe, Peter M. Howley, Diane E. Griffin, Robert A. Lamb, Malcolm A. Martin, Bernard Roizman, Stephen E. Straus. 7th edition, 2020

Reference Books:

1. Principles of Molecular Virology – Alan J Cann. 7th edition 2023
2. Virology: Principles and Applications – J Carter, V Saunders, 2nd edition, 2013
3. **Knipe, D. M., & Howley, P. M. (Eds.).** (2020). *Fields virology* (7th ed.). Lippincott Williams & Wilkins.
4. **Flint, S. J., Racaniello, V. R., Rall, G. F., Skalka, A. M., & Enquist, L. W.** (2020). *Principles of virology* (5th ed.). ASM Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Group discussion Flipped classroom

Subject name: Neurobiochemistry		
Subject code:		
Credit: 3-1-0-4		
SCHEME OF EVALUATION: THEORY PAPER (T)		
Course objectives		
This paper provides a basic understanding of the nervous system and nerve cell functioning and diseases associated with the nervous system		
Course Outcome		
On Successful Completion of the course the students will be able to:		
Sl No	Course Outcome	Blooms

		Taxonomy Level
CO 1	Recognize the mechanism of nervous system function, transmissions, transmitters and neuro-degradative diseases.	BT1
CO 2	Distinguish the biochemistry of nerve signaling, transmitters and neuro-degradative diseases.	BT2
CO 3	Analyse the functioning of the nervous systems and the significance the various transmitters in the nerve functioning.	BT3
CO4	Distinguish the the biochemical events leading to the development of different neuro-degradative diseases.	BT4

Course contents:

Module	Topic	Period
1	Neuron: Neuro cellular anatomy, neural membrane, classification of neuron, nerve fibers, axonal transport, neural growth, neuroglia, nervous system, blood-brain barrier, cerebrospinal fluid.	15
2	Neuronal signalling: Membrane potentials, ion channels, recording neuronal signals, ionic basis of resting potential and action potential, propagation of action potential. Synaptic transmission- Synapse, Electrical synapse transmission, chemical synaptic transmission, Synaptic transmitter release, synaptic potentials, synaptic delay, synaptic plasticity, molecular mechanism of synaptic transmission, myoneural junction.	15
3	Neurotransmitters: Chemistry, synthesis, storage, release, receptors and function- acetyl choline, catecholamines, serotonin, histamine, glutamate, asparatate, GABA, glycine, neuropeptides, nitric oxide.	15
4	Neural processing and neurodegenerative disorders: Learning and memory, neurochemical basis of drug abuse, neurodegenerative disorders, Parkinson's disorder, Alzheimer's disorder, Amyotrophic Lateral Sclerosis, Senile Dementia	15

TOTAL		60
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Text Books:

1. Arthur C. Guyton and John E Hall, Text book of medical physiology 11th Edition;2006
2. David Nelson and Michael Cox, Lehninger Principles of Biochemistry, 4th edition;2005

Reference books:

1. Bruce Alberts, Alexander Johnson, Juliana Lewis, Martin Raff, Keith Roberts and Peter Walter, Molecular biology of the cell, 4th Edition;2004
2. Gordon Shepherd, Neurobiology, 3rd Edition;1994
3. Mark F Bear, Barry W Connors and Michael A Paradiso, Neuroscience: Exploring the brain, 4th Edition; 2015
4. Siegel, G. J., Albers, R. W., Brady, S. T., & Price, D. L. (Eds.). (2006). *Basic neurochemistry: Molecular, cellular, and medical aspects* (7th ed.). Elsevier Academic Press.

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Group discussion Flipped classroom

Subject name: Molecular Endocrinology Subject code: Credit :3-1-0-4
Course Objective
The objectives of this course are to introduce the basic principles, organs and systems in mammalian (human) endocrinology

Course Outcome		
On Successful Completion of the course the students will be able to:		
Sl No	Course Outcome	Blooms Taxonomy Level
CO 1	Recognize the endocrine glands and their diseases	BT1
CO 2	Distinguish between the workings of the different hormones and the resultant diseases	BT2
CO 3	Analyse the biochemistry of hormone actions and the responsible proteins involved in regulating hormone action	BT3
CO4	Evaluate the reasons and mode of action of various drugs aimed at countering hormonal diseases..	BT4

Course contents:

Module	Topic	Period
1	Hypothalamo - Hypophysial – Hormones: Bio-chemistry and mechanism of action, regulation, synthesis and secretion Hypo and hyperactivity of pituitary hormones - Hypothalamic releasing factors, Anterior Pituitary hormones, Vasopressin, Oxytocin. Regulation of synthesis. Lactogenic hormones. Glycoprotein hormones of the POMC family, endorphins, Gigantism, Dwarfism, Acromegaly, diabetic insipidus, syndromes of inappropriate ADH secretion. Mechanism of action and functions of melatonin	15
2	Thyroid and para thyroid hormones: Synthesis, secretion, transport and mechanism of action. Metabolic fate and biological action, thyroid diseases, thyrotoxicosis, goiter, hypothyroidism, grave's disease, Hashimoto's disease, thyroid function tests, calcium and phosphorus metabolism, calcitriol, pathophysiology.	15
3	Adrenal hormones: Adrenal cortex, glucocorticoids and mineralocorticoids, Synthesis, secretion, transport and mechanism of action. Metabolic fate and biological actions, Hormones of Adrenal Medulla - Catecholamines - Biosynthesis, storage, metabolism, regulation of synthesis. Abnormal secretion of adrenal	15

	hormones, congenital adrenal, hyperplasia, phaeocromocytoma. Gonadal hormones, Androgen, estrogen, synthesis, secretion, transport and mechanism of action. Metabolic fate and biological action, Ovarian cycle. Pregnancy, Biochemical changes in pregnancy.	
4	Pancreatic Hormones: Islets of langerhans and hormone secretions, biosynthesis, secretion and mechanism of action, receptor signaling, pathway of insulin and glucagon, somatostatin, pancreatic poly peptide and insulin- like growth factors, gastro intestinal hormones- synthesis, structure, secretion and function. GIP, VIP, gastric, CCK and other peptides.	15
TOTAL		60

Text books-

1. Hadley, M. E. (2006). *Endocrinology*. Pearson Education India.
2. Melmed, S., Polonsky, K. S., Larsen, P. R., & Kronenberg, H. M. (2015). *Williams textbook of endocrinology E-Book*. Elsevier Health Sciences.

Reference books-

1. Lehninger Principles of Biochemistry, Nelson, D.L., Cox, M.M. 7th Edition, 2017, WH Freeman and Company, New York, USA
2. Biochemistry; Donald Voet, Judith G. Voet, 4th Edition, John Wiley and sons (2010) .
3. Devlin T. M. (2011) Textbook of Biochemistry with clinical correlations, Wiley-Liss Publ.
4. Hall, J.E. (2015), Guyton and Hall Textbook of Medical Physiology, Elsevier

Credit Distribution		
Lecture/Tutorial	Practical	Experiential Learning
60 hours	0 hours	30 hours Assignment Group discussion Flipped classroom